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IEEE Standard for Ethernet

Amendment 3: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Operation over Optical Fiber Using 100 Gb/s Signaling

STANDARDS

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.3db™-2022
(Amendment to IEEE Std 802.3™-2022
as amended by IEEE Std 802.3dd™-2022
and IEEE Std 802.3cs™-2022)

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Approved 21 September 2022
IEEE SA Standards Board

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Abstract: This amendment to IEEE Std 802.3-2022 adds Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Ethernet optical interfaces based on 100 Gb/s per wavelength optical signaling over multimode fiber.

Keywords: 100 Gb/s Ethernet, 100GBASE-SR1, 100GBASE-VR1, 200 Gb/s Ethernet, 200GBASE-SR2, 200GBASE-VR2, 400 Gb/s Ethernet, 400GBASE-SR4, 400GBASE-VR4, amendment, Energy-Efficient Ethernet (EEE), Ethernet, forward error correction (FEC), IEEE 802.3™, IEEE 802.3cs™, IEEE 802.3db™, IEEE 802.3dd™, multimode fiber (MMF), PAM4, Physical Medium Dependent (PMD) sublayer

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Adam Healey, *IEEE 802.3 Working Group Vice-Chair*
Jon Lewis, *IEEE 802.3 Working Group Secretary*
Steven B. Carlson, *IEEE 802.3 Working Group Executive Secretary*
Valerie Maguire, *IEEE 802.3 Working Group Treasurer*

Robert Lingle, *IEEE P802.3db 100 Gb/s, 200 Gb/s, and 400 Gb/s Short Reach Fiber Task Force Chair*
Mabud Choudhury, *IEEE P802.3db 100 Gb/s, 200 Gb/s, and 400 Gb/s Short Reach Fiber Task Force Vice-Chair*
Ramana Murty, *IEEE P802.3db 100 Gb/s, 200 Gb/s, and 400 Gb/s Short Reach Fiber Task Force Co-Editor*
Earl Parsons, *IEEE P802.3db 100 Gb/s, 200 Gb/s, and 400 Gb/s Short Reach Fiber Task Force Co-Editor*

John Abbott
Rob Aekins
Stefan Andrae
Pete Anslow
Michikazu Aono
Nobuyasu Araki
Joseph Aronson
Tim Baggett
Thananya Baldwin
Fernando Barbero
Denis Beaudoin
Liav Ben-Artsi
Vipul Bhatt
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Ichiro Ogura
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Introduction

This introduction is not part of IEEE Std 802.3db-2022, IEEE Standard for Ethernet—Amendment 3: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Operation over Optical Fiber using 100 Gb/s Signaling.

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel, and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an Ethertype to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called *Fast Ethernet*), IEEE Std 802.3z™ added 1000 Mb/s operation (also called *Gigabit Ethernet*), IEEE Std 802.3ae™ added 10 Gb/s operation (also called *10 Gigabit Ethernet*), IEEE Std 802.3ah™ specified access network Ethernet (also called *Ethernet in the First Mile*), and IEEE Std 802.3ba™ added 40 Gb/s operation (also called *40 Gigabit Ethernet*) and 100 Gb/s operation (also called *100 Gigabit Ethernet*). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of IEEE Std 802.3db-2022 publication, IEEE Std 802.3 was composed of the following documents:

IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex H and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats, and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33E. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

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Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well the 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

IEEE Std 802.3cs™-2022

Amendment 2—This amendment to IEEE Std 802.3-2022 defines Super-PON optical subscriber access networks, in the family of Ethernet passive optical networks (EPONs). Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN)

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through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This amendment specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Media Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

IEEE Std 802.3db™-2022

Amendment 3—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 167. This amendment adds Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s over one, two, and four pairs of multimode fiber based on 100 Gb/s optical signaling.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

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IEEE Standard for Ethernet

Amendment 3: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Operation over Optical Fiber Using 100 Gb/s Signaling

(This amendment is based on IEEE Std 802.3-2022 as amended by IEEE Std 802.3dd-2022 and IEEE Std 802.3cs-2022.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.[6]

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strikethrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in green.

[6] Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

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1. Introduction

1.3 Normative references

Insert the following reference alphanumerically in 1.3:

IEC 63267-1, Fibre optic interconnecting devices and passive components—Connector optical interfaces for enhanced macro bend loss multimode fibres—Part 1: Optical interfaces for 50 um core diameter fibres—General and guidance.

1.4 Definitions

Insert the following new definition after 1.4.38 "100GBASE-R encoding":

1.4.38a 100GBASE-SR1: IEEE 802.3 Physical Layer specification for 100 Gb/s using 100GBASE-R encoding and 4-level pulse amplitude modulation over multimode fiber, with reach up to at least 100 m. (See IEEE Std 802.3, Clause 167.)

Insert the following new definition after 1.4.41 "100GBASE-SR4":

1.4.41a 100GBASE-VR1: IEEE 802.3 Physical Layer specification for 100 Gb/s using 100GBASE-R encoding and 4-level pulse amplitude modulation over multimode fiber, with reach up to at least 50 m. (See IEEE Std 802.3, Clause 167.)

Insert the following new definition after 1.4.108 "200GBASE-R":

1.4.108a 200GBASE-SR2: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding and 4-level pulse amplitude modulation over two lanes of multimode fiber, with reach up to at least 100 m. (See IEEE Std 802.3, Clause 167.)

Insert the following new definition after 1.4.109 "200GBASE-SR4":

1.4.109a 200GBASE-VR2: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding and 4-level pulse amplitude modulation over two lanes of multimode fiber, with reach up to at least 50 m. (See IEEE Std 802.3, Clause 167.)

Insert the following new definition after 1.4.142 "400GBASE-SR16":

1.4.142a 400GBASE-SR4: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding and 4-level pulse amplitude modulation over four lanes of multimode fiber, with reach up to at least 100 m. (See IEEE Std 802.3, Clause 1.)

Insert the following new definition after 1.4.144 "400GBASE-SR8":

1.4.144a 400GBASE-VR4: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding and 4-level pulse amplitude modulation over four lanes of multimode fiber, with reach up to at least 50 m. (See IEEE Std 802.3, Clause 167.)

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30. Management

30.5 Layer management for medium attachment units (MAUs)

30.5.1 MAU managed object class

30.5.1.1 MAU attributes

30.5.1.1.2 aMAUType

ATTRIBUTE

APPROPRIATE SYNTAX:

Insert the following new entry into "APPROPRIATE SYNTAX" in 30.5.1.1.2 after 100GBASE-R as follows:

100GBASE-SR1	100GBASE-R PCS/PMA over multimode fiber PMD with reach up to at least 100 m as specified in Clause 167

Insert the following new entry into "APPROPRIATE SYNTAX" in 30.5.1.1.2 after 100GBASE-SR10 as follows:

100GBASE-VR1	100GBASE-R PCS/PMA over multimode fiber PMD with reach up to at least 50 m as specified in Clause 167

Insert the following new entry into "APPROPRIATE SYNTAX" in 30.5.1.1.2 after 200GBASE-R as follows:

200GBASE-SR2	200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with reach up to at least 100 m as specified in Clause 167

Insert the following new entry into "APPROPRIATE SYNTAX" in 30.5.1.1.2 after 200GBASE-SR4 as follows:

200GBASE-VR2	200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with reach up to at least 50 m as specified in Clause 167

Insert the following new entry into "APPROPRIATE SYNTAX" in 30.5.1.1.2 after 400GBASE-R as follows:

400GBASE-SR4	400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with reach up to at least 100 m as specified in Clause 167

Insert the following new entry into "APPROPRIATE SYNTAX" in 30.5.1.1.2 after 400GBASE-SR16 as follows:

400GBASE-VR4	400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with reach up to at least 50 m as specified in Clause 167

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45. Management Data Input/Output (MDIO) Interface

45.2 MDIO Interface registers

45.2.1 PMA/PMD registers

45.2.1.6 PMA/PMD control 2 register (Register 1.7)

Change the description of bits 1.7.6:0 in Table 45–7 as shown (unchanged rows not shown):

Table 45–7—PMA/PMD control 2 register bit definitions

Bit(s)	Name	Description	R/W[a]
…			
1.7.6:0	PMA/PMD type selection	6 5 4 3 2 1 0 1 1 1 1 1 1 1 = reserved 1 1 1 1 1 1 0 = 400GBASE-SR4 PMA/PMD 1 1 1 1 1 0 1 = 400GBASE-VR4 PMA/PMD 1 1 1 1 1 0 0 = 200GBASE-SR2 PMA/PMD 1 1 1 1 0 1 1 = 200GBASE-VR2 PMA/PMD 1 1 1 1 0 1 0 = 100GBASE-SR1 PMA/PMD 1 1 1 1 0 0 1 = 100GBASE-VR1 PMA/PMD ~~1 1 1 1 1 x x = reserved~~ ~~1 1 1 1 0 1 x = reserved~~ ~~1 1 1 1 0 0 1 = reserved~~ …	R/W

[a]R/W = Read/Write, RO = Read only

45.2.1.7 PMA/PMD status 2 register (Register 1.8)

45.2.1.7.4 Transmit fault (1.8.11)

Insert a new row into Table 45–9 after the 100GBASE-CR4 row:

Table 45–9—Transmit fault description location

PMA/PMD	Description location
100GBASE-VR1, 100GBASE-SR1, 200GBASE-VR2, 200GBASE-SR2, 400GBASE-VR4, 400GBASE-SR4	167.5.10

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45.2.1.7.5 Receive fault (1.8.10)

Insert a new row into Table 45–10 after the 100GBASE-CR4 row:

Table 45–10—Receive fault description location

PMA/PMD	Description location
100GBASE-VR1, 100GBASE-SR1, 200GBASE-VR2, 200GBASE-SR2, 400GBASE-VR4, 400GBASE-SR4	167.5.11

45.2.1.8 PMD transmit disable register (Register 1.9)

Insert a new row into Table 45–12 after the 100GBASE-CR4 row:

Table 45–12—Transmit disable description location

PMA/PMD	Description location
100GBASE-VR1, 100GBASE-SR1, 200GBASE-VR2, 200GBASE-SR2, 400GBASE-VR4, and 400GBASE-SR4	167.5.7

45.2.1.21 200G PMA/PMD extended ability register (Register 1.23)

Change the row for bits 1.23.14:7 in Table 45–24 as follows (unchanged rows not shown):

Table 45–24—200G PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W[a]
…			
1.23.14:11~~7~~	Reserved	Value always 0	RO
1.23.10	200GBASE-SR2 ability	1 = PMA/PMD is able to perform 200GBASE-SR2 0 = PMA/PMD is not able to perform 200GBASE-SR2	RO
1.23.9	200GBASE-VR2 ability	1 = PMA/PMD is able to perform 200GBASE-VR2 0 = PMA/PMD is not able to perform 200GBASE-VR2	RO
1.23.8:7	Reserved	Value always 0	RO
…			

[a]RO = read only

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Insert the following new subclauses (45.2.1.21.1a and 45.2.1.21.1b) after 45.2.1.21.1:

45.2.1.21.1a 200GBASE-SR2 ability (1.23.10)

When read as a one, bit 1.23.10 indicates that the PMA/PMD is able to operate as a 200GBASE-SR2 PMA/PMD type. When read as a zero, bit 1.23.10 indicates that the PMA/PMD is not able to operate as a 200GBASE-SR2 PMA/PMD type.

45.2.1.21.1b 200GBASE-VR2 ability (1.23.9)

When read as a one, bit 1.23.9 indicates that the PMA/PMD is able to operate as a 200GBASE-VR2 PMA/PMD type. When read as a zero, bit 1.23.9 indicates that the PMA/PMD is not able to operate as a 200GBASE-VR2 PMA/PMD type.

45.2.1.22 400G PMA/PMD extended ability register (Register 1.24)

Change the row for bits 1.24.14:11 in Table 45–25 as follows (some unchanged rows not shown):

Table 45–25—400G PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W[a]
…			
1.24.14~~:11~~	Reserved	Value always 0	RO
1.24.13	400GBASE-SR4 ability	1 = PMA/PMD is able to perform 400GBASE-SR4 0 = PMA/PMD is not able to perform 400GBASE-SR4	RO
1.24.12	400GBASE-VR4 ability	1 = PMA/PMD is able to perform 400GBASE-VR4 0 = PMA/PMD is not able to perform 400GBASE-VR4	RO
1.24.11	Reserved	Value always 0	RO
…			

[a]RO = read only

Insert the following new subclauses (45.2.1.22.1a and 45.2.1.22.1b) after 45.2.1.22.1:

45.2.1.22.1a 400GBASE-SR4 ability (1.24.13)

When read as a one, bit 1.24.13 indicates that the PMA/PMD is able to operate as a 400GBASE-SR4 PMA/PMD type. When read as a zero, bit 1.24.13 indicates that the PMA/PMD is not able to operate as a 400GBASE-SR4 PMA/PMD type.

45.2.1.22.1b 400GBASE-VR4 ability (1.24.12)

When read as a one, bit 1.24.12 indicates that the PMA/PMD is able to operate as a 400GBASE-VR4 PMA/PMD type. When read as a zero, bit 1.24.12 indicates that the PMA/PMD is not able to operate as a 400GBASE-VR4 PMA/PMD type.

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45.2.1.24 40G/100G PMA/PMD extended ability 2 (Register 1.26)

Change the rows for bits 1.26.2:0 and 1.26.15:10 in Table 45–27 as follows (some unchanged rows not shown):

Table 45–27—40G/100G PMA/PMD extended ability 2 register bit definitions

Bit(s)	Name	Description	R/W[a]
1.26.15:11~~10~~	Reserved	Value always 0	RO
1.26.10	100GBASE-VR1 ability	1 = PMA/PMD is able to perform 100GBASE-VR1 0 = PMA/PMD is not able to perform 100GBASE-VR1	RO
…			
1.26.2	100GBASE-SR1 ability	1 = PMA/PMD is able to perform 100GBASE-SR1 0 = PMA/PMD is not able to perform 100GBASE-SR1	RO
1.26.~~2~~1:0	Reserved	Value always 0	RO

[a]RO = read only

Insert the following new subclause 45.2.1.24a after 45.2.1.24, below Table 45-27:

45.2.1.24a 100GBASE-VR1 ability (1.26.10)

When read as a one, bit 1.26.10 indicates that the PMA/PMD is able to operate as a 100GBASE-VR1 PMA/PMD type. When read as a zero, bit 1.26.10 indicates that the PMA/PMD is not able to operate as a 100GBASE-VR1 PMA/PMD type.

Insert the following new subclause 45.2.1.24.8 after 45.2.1.24.7:

45.2.1.24.8 100GBASE-SR1 ability (1.26.2)

When read as a one, bit 1.26.2 indicates that the PMA/PMD is able to operate as a 100GBASE-SR1 PMA/PMD type. When read as a zero, bit 1.26.2 indicates that the PMA/PMD is not able to operate as a 100GBASE-SR1 PMA/PMD type.

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78. Energy-Efficient Ethernet (EEE)

78.1 Overview

78.1.4 PHY types optionally supporting EEE

Insert new rows for 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 in Table 78–1 with 100GBASE-VR1 after 100GBASE-CR10, 200GBASE-VR2 after 200GBASE-CR4, 400GBASE-VR4 after 200GBASE-ER4, 100GBASE-SR1 after 100GBASE-SR2, 200GBASE-SR2 after 200GBASE-SR4, and 400GBASE-SR4 after 400GBASE-SR8 as follows (some unchanged rows not shown):

Table 78–1—Clauses associated with each PHY or interface type

PHY or interface type	Clause
...	
100GBASE-CR10	82, 83, 85, 74
100GBASE-VR1[b]	82, 135, 167
100GBASE-SR4[b]	82, 83, 91, 95
100GBASE-SR2[b]	82, 135, 138
100GBASE-SR1[b]	82, 135, 167
100GBASE-SR10[b]	82, 83, 86
100GBASE-DR[b]	82, 135, 140
...	
200GBASE-CR4[b]	119, 120, 137
200GBASE-VR2[b]	119, 120, 167
200GBASE-SR4[b]	119, 120, 138
200GBASE-SR2[b]	119, 120, 167
200GBASE-DR4[b]	119, 120, 121
...	
200GBASE-ER4[b]	119, 120, 122
400GBASE-VR4[b]	119, 120, 167
400GBASE-SR16[b]	119, 120, 123
400GBASE-SR8[b]	119, 120, 138
400GBASE-SR4[b]	119, 120, 167
400GBASE-SR4.2[b]	119, 120, 150
400GBASE-DR4[b]	119, 120, 124
...	

[b] The deep sleep mode of EEE is not supported for this PHY.

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80. Introduction to 40 Gb/s and 100 Gb/s networks

80.1 Overview

80.1.3 Relationship of 40 Gigabit and 100 Gigabit Ethernet to the ISO OSI reference model

Change list item h) in 80.1.3 as follows:

h) The MDIs as specified in:
- Clause 89 for 40GBASE-FR
- Clause 140 for 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1
- Clause 154 for 100GBASE-ZR
- Clause 167 for 100GBASE-VR1 and 100GBASE-SR1

use a single lane data path.

80.1.4 Nomenclature

Change Table 80–1 as follows (some unchanged rows not shown):

Table 80–1—40 Gb/s and 100 Gb/s PHYs

Name	Description
...	
100GBASE-CR10	100 Gb/s PHY using 100GBASE-R encoding over ten lanes of shielded balanced copper cabling, with reach up to at least 7 m (see Clause 85)
100GBASE-VR1	100 Gb/s PHY using 100GBASE-R encoding over multimode fiber, with reach up to at least 50 m (see Clause 167)
100GBASE-SR4	100 Gb/s PHY using 100GBASE-R encoding over four lanes of multimode fiber, with reach up to at least 100 m (see Clause 95)
100GBASE-SR2	100 Gb/s PHY using 100GBASE-R encoding over two lanes of multimode fiber, with reach up to at least 100 m (see Clause 138)
100GBASE-SR1	100 Gb/s PHY using 100GBASE-R encoding over multimode fiber, with reach up to at least 100 m (see Clause 167)
100GBASE-SR10	100 Gb/s PHY using 100GBASE-R encoding over ten lanes of multimode fiber, with reach up to at least 100 m (see Clause 86)
100GBASE-DR	100 Gb/s PHY using 100GBASE-R encoding over single-mode fiber, with reach up to at least 500 m (see Clause 140)
...	

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80.1.5 Physical Layer signaling systems

Change Table 80–5 as follows:

Table 80–5—Nomenclature and clause correlation (100GBASE-P optical)

Nomenclature	Clause[a] 78	81		82	83	83A	83B	83D	83E	91	135	135D	135E	135F	135G	138	140			167	
	EEE	RS	CGMII	PCS	100GBASE-R PMA	CAUI-10	CAUI-10	CAUI-4	CAUI-4	RS-FEC	100GBASE-P PMA	100GAUI-4 C2C	100GAUI-4 C2M	100GAUI-2 C2C	100GAUI-2 C2M	100GBASE-SR2	100GBASE-DR	100GBASE-FR1	100GBASE-LR1	100GBASE-VR1	100GBASE-SR1
100GBASE-VR1	O	M	O	M	O	O	O	O	O	M	M	O	O	O	O					M	
100GBASE-SR2	O	M	O	M	O	O	O	O	O	M	M	O	O	O	O	M					
100GBASE-SR1	O	M	O	M	O	O	O	O	O	M	M	O	O	O	O						M
100GBASE-DR	O	M	O	M	O	O	O	O	O	M	M	O	O	O	O		M				
100GBASE-FR1	O	M	O	M	O	O	O	O	O	M	M	O	O	O	O			M			
100GBASE-LR1	O	M	O	M	O	O	O	O	O	M	M	O	O	O	O				M		

[a]O = Optional, M = Mandatory.

80.2 Summary of 40 Gigabit and 100 Gigabit Ethernet sublayers

Change the first paragraph in 80.2.3 as follows:

80.2.3 Forward error correction (FEC) sublayers

A forward error correction sublayer is optional for 40GBASE-KR4, 40GBASE-CR4, and 100GBASE-CR10 PHYs and mandatory for 100GBASE-CR2, 100GBASE-CR4, 100GBASE-KR2, 100GBASE-KR4, 100GBASE-KP4, 100GBASE-VR1, 100GBASE-SR1, 100GBASE-SR2, 100GBASE-SR4, 100GBASE-DR, 100GBASE-FR1, 100GBASE-LR1, and 100GBASE-ZR PHYs. The FEC sublayer can be placed in between the PCS and PMA sublayers or between two PMA sublayers.

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80.4 Delay constraints

Change Table 80–7 as follows (some unchanged rows not shown):

Table 80–7—Sublayer delay constraints

Sublayer	Maximum (bit time)[a]	Maximum (pause_quanta)[b]	Maximum (ns)	Notes[c]
...				
100GBASE-CR10 PMD	9 728	19	97.28	Does not include delay of cable medium. See 92.4.
100GBASE-VR1 PMD	2 048	4	20.48	Includes 2 m of fiber. See 167.3.
100GBASE-SR4 PMD	2 048	4	20.48	Includes 2 m of fiber. See 95.3.1.
100GBASE-SR2 PMD	2 048	4	20.48	Includes 2 m of fiber. See 138.3.
100GBASE-SR1 PMD	2 048	4	20.48	Includes 2 m of fiber. See 167.3.
100GBASE-SR10 PMD	2 048	4	20.48	Includes 2 m of fiber. See 86.3.1.
...				

[a] For 40GBASE-R, 1 bit time (BT) is equal to 25 ps and for 100GBASE-R, 1 bit time (BT) is equal to 10 ps. (See 1.4.215 for the definition of bit time.)
[b] For 40GBASE-R, 1 pause_quantum is equal to 12.8 ns and for 100GBASE-R, 1 pause_quantum is equal to 5.12 ns. (See 31B.2 for the definition of pause_quanta.)
[c] Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

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80.5 Skew constraints

Change Table 80-8 and Table 80-9 as follows:

Table 80–8—Summary of Skew constraints

Skew points	Maximum Skew (ns)[a]	Maximum Skew for 40GBASE-R PCS lane (UI)[b]	Maximum Skew for 100GBASE-R PCS lane (UI)[c]	Notes[d]
SP0	29	N/A	≈ 150	See 83.5.3.1 or 135.5.3
SP1	29	≈ 299	≈ 150	See 83.5.3.2 or 135.5.3
SP2	43	≈ 443	≈ 222	See 83.5.3.4, 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 140.3, ~~or~~ 154.3.2, or 167.3.2
SP3	54	≈ 557	≈ 278	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 140.3, ~~or~~ 154.3.2, or 167.3.2
SP4	134	≈ 1382	≈ 691	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2 135.5.3, 136.6, 137.6, 138.3, 140.3, ~~or~~ 154.3.2, or 167.3.2
SP5	145	≈ 1495	≈ 748	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 140.3, ~~or~~ 154.3.2, or 167.3.2
SP6	160	≈ 1649	≈ 824	See 83.5.3.6 or 135.5.3
SP7	29	N/A	≈ 150	See 83.5.3.8 or 135.5.3
At PCS receive	180	≈ 1856	≈ 928	See 82.2.13
At RS-FEC transmit	49	N/A	≈ 253	See 91.5.2.2
At RS-FEC receive[e]	180	N/A	≈ 4641	See 91.5.3.1
At PCS receive (with RS-FEC)	49	N/A	≈ 253	See 82.2.13

[a]The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.
[b]The symbol ≈ indicates approximate equivalent of maximum Skew in UI for 40GBASE-R, based on 1 UI equals 96.969697 ps at PCS lane signaling rate of 10.3125 GBd.
[c]The symbol ≈ indicates approximate equivalent of maximum Skew in UI for 100GBASE-R, based on 1 UI equals 193.939394 ps at PCS lane signaling rate of 5.15625 GBd.
[d]Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.
[e]The skew at the RS-FEC receive is the skew between RS-FEC lanes. The symbol ≈ indicates approximate equivalent of maximum Skew in UI for RS-FEC lanes with a signaling rate of 25.78125 GBd.

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Table 80–9—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation				Notes[a]
	ns	UI for 10.3125 GBd lane[b]	UI for 25.78125 GBd lane[c]	UI for 26.5625 GBd lane[d]	
SP0	0.2	≈ 2	N/A	N/A	See 83.5.3.1 or 135.5.3
SP1	0.2	≈ 2	N/A	≈ 5	See 83.5.3.2 or 135.5.3
SP2	0.4	≈ 4	≈ 10	≈ 11	See 83.5.3.4, 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, ~~or~~ 138.3, or 167.3
SP3	0.6	≈ 6	≈ 15	≈ 16	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 136.6, 137.6, ~~or~~ 138.3, or 167.3
SP4	3.4	≈ 35	≈ 88	≈ 90	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, or 95.3.2, 136.6, 137.6, ~~or~~ 138.3, or 167.3
SP5	3.6	≈ 37	≈ 93	≈ 96	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, or 167.3
SP6	3.8	≈ 39	≈ 98	≈ 101	See 83.5.3.6 or 135.5.3
SP7	0.2	≈ 2	N/A	N/A	See 83.5.3.8 or 135.5.3
At PCS receive	4	≈ 41	N/A	N/A	See 82.2.13
At RS-FEC transmit	0.4	N/A	≈ 10	≈ 11	See 91.5.2.2
At RS-FEC receive[e]	4	N/A	≈ 103	≈ 106	See 91.5.3.1
At PCS receive (with RS-FEC)	0.4	N/A	≈ 10	N/A	See 82.2.13

[a]Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.
[b]The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI for 40GBASE-R, based on 1 UI equals 96.969697 ps at PMD lane signaling rate of 10.3125 GBd.
[c]The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI for 100GBASE-R, based on 1 UI equals 38.787879 ps at PMD lane signaling rate of 25.78125 GBd.
[d]The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI for 100GBASE-R, based on 1 UI equals 37.64706 ps at PMD lane signaling rate of 26.5625 GBd.
[e]The skew at the RS-FEC receive is the skew between RS-FEC lanes.

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91. Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs

91.5 Functions within the RS-FEC sublayer

91.5.2.7 Reed-Solomon encoder

Change the second sentence of the second paragraph of 91.5.2.7 as follows:

When used to form a 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1 PHY, the RS-FEC sublayer shall implement RS(544,514).

91.5.3 Receive function

91.5.3.3 Reed-Solomon decoder

Change the second sentence of the second paragraph of 91.5.3.3 as follows:

When used to form a 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1 PHY, the RS-FEC sublayer shall be capable of correcting any combination of up to t=15 symbol errors in a codeword.

Change the third paragraph of 91.5.3.3 as follows:

The Reed-Solomon decoder may provide the option to perform error detection without error correction to reduce the delay contributed by the RS-FEC sublayer. The presence of this option is indicated by the assertion of the FEC_bypass_correction_ability variable (see 91.6.8). When the option is provided, it is enabled by the assertion of the FEC_bypass_correction_enable variable (see 91.6.1). This option shall not be used when the RS-FEC sublayer is used to form part of a 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR1, 100GBASE-SR2, 100GBASE-SR4, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1 PHY.

Change the last sentence of the last paragraph of 91.5.3.3 as follows:

When the RS-FEC sublayer is used to form a 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1 PHY, the symbol error threshold shall be K=6380.

91.5.3.3.1 FEC Degraded SER (optional)

Change the first paragraph of 91.5.3.3.1 as follows:

For 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1 PHYs an optional FEC degraded symbol error ratio function is available.

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91.6 RS-FEC MDIO function mapping

Change 91.6.3 as follows:

91.6.3 four_lane_pmd

When this variable is set to one, the alignment marker mapping function substitutes the fixed bytes of the alignment markers corresponding to PCS lanes 17, 18, and 19 with the fixed bytes for the alignment marker corresponding to PCS lane 16 (see 91.5.2.6). When this variable is set to zero, the alignment markers corresponding to PCS lanes 17, 18, and 19 are passed through unmodified. The default value of the variable is one which is the value required by the 100GBASE-CR4, 100GBASE-KR4, 100GBASE-KP4, and 100GBASE-SR4 PMDs. This variable is set to zero for the 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1 PMDs. This variable is mapped to the bit defined in 45.2.1.116 (1.200.3).

91.7 Protocol implementation conformance statement (PICS) proforma for Clause 91, Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs[7]

Change the table in 91.7.3 as follows (some unchanged rows not shown):

91.7.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
...					
*KP4	100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1		Used to form complete 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1 PHY	O	Yes [ ] No [ ]
*FDDP	100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1	91.5.3.3.1	Used to form complete 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1 PHY	O	Yes [ ] No [ ]
...					

[7]*Copyright release for PICS proformas*: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

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91.7.4 PICS proforma tables for Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs

91.7.4.1 Transmit function

Change the table in 91.7.4.1 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
TF11	Reed-Solomon encoder for 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1	91.5.2.7	RS(544,514)	KP4:M	Yes [ ] N/A [ ]
...					

91.7.4.2 Receive function

Change the table in 91.7.4.2 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
RF4	Reed-Solomon decoder for 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1	91.5.3.3	Corrects any combination of up to t=15 symbol errors in a codeword unless error correction bypassed	KP4:M	Yes [ ] N/A [ ]
...					
RF6	Error correction bypass for 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-SR4, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1	91.5.3.3	Error correction is not bypassed	SR4:M or FDDP:M	Yes [ ] N/A [ ]

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Item	Feature	Subclause	Value/Comment	Status	Support
...					
RF12	Symbol error threshold for 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1	91.5.3.3	*K*=6380	BEI* KP4:M	Yes [ ] N/A [ ]
...					

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116. Introduction to 200 Gb/s and 400 Gb/s networks

116.1 Overview

116.1.2 Relationship of 200 Gigabit and 400 Gigabit Ethernet to the ISO OSI reference model

Change item h) in the lettered list in 116.1.2 and insert item i) at the end of the lettered list in 116.1.2 as follows:

h) The MDIs as specified in:
   — Clause 121 for 200GBASE-DR4
   — Clause 122 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4
   — Clause 124 for 400GBASE-DR4
   — Clause 136 for 200GBASE-CR4
   — Clause 137 for 200GBASE-KR4
   — Clause 138 for 200GBASE-SR4
   — Clause 151 for 400GBASE-FR4 and 400GBASE-LR4-6
   — Clause 167 for 400GBASE-VR4 and 400GBASE-SR4
   all use a 4-lane data path.
i) The MDIs as specified in Clause 167 for 200GBASE-VR2 and 200GBASE-SR2 each use a 2-lane data path.

116.1.3 Nomenclature

Change Table 116–1 as follows (some unchanged rows not shown):

Table 116–1—200 Gb/s PHYs

Name	Description
...	
200GBASE-CR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of twinaxial copper cable (see 1.4.559 and Clause 136)
200GBASE-VR2	200 Gb/s PHY using 200GBASE-R encoding over two lanes of multimode fiber, with reach up to at least 50 m (see Clause 167)
200GBASE-SR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of multimode fiber (see Clause 138)
200GBASE-SR2	200 Gb/s PHY using 200GBASE-R encoding over two lanes of multimode fiber, with reach up to at least 100 m (see Clause 167)
200GBASE-DR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of single-mode fiber, with reach up to at least 500 m (see Clause 121)
...	

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Change Table 116–2 as follows (some unchanged rows not shown):

Table 116–2—400 Gb/s PHYs

Name	Description
<u>400GBASE-VR4</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over four lanes of multimode fiber, with reach up to at least 50 m (see Clause 167)</u>
400GBASE-SR16	400 Gb/s PHY using 400GBASE-R encoding over sixteen lanes of multimode fiber, with reach up to at least 100 m (see Clause 123)
400GBASE-SR8	400 Gb/s PHY using 400GBASE-R encoding over eight lanes of multimode fiber, with reach up to at least 100 m (see Clause 138)
<u>400GBASE-SR4</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over four lanes of multimode fiber, with reach up to at least 100 m (see Clause 167)</u>
400GBASE-SR4.2	400 Gb/s PHY using 400GBASE-R encoding over eight lanes on multimode fiber in a bidirectional WDM format, with reach up to at least 150 m (see Clause 150)
...	

116.1.4 Physical Layer signaling systems

Change Table 116–4 as follows:

Table 116–4—PHY type and clause correlation (200GBASE optical)

PHY type	Clause[a]																
	78	117		118	119	120	120B	120C	120D	120E	121	122			138	<u>167</u>	
	EEE	RS	200GMII	200GMII Extender	200GBASE-R PCS	200GBASE-R PMA	200GAUI-8 C2C	200GAUI-8 C2M	200GAUI-4 C2C	200GAUI-4 C2M	200GBASE-DR4 PMD	200GBASE-FR4 PMD	200GBASE-LR4 PMD	200GBASE-ER4 PMD	200GBASE-SR4 PMD	<u>200GBASE-VR2 PMD</u>	<u>200GBASE-SR2 PMD</u>
<u>200GBASE-VR2</u>	<u>O</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>M</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>O</u>	<u>O</u>						<u>M</u>	
200GBASE-SR4	O	M	O	O	M	M	O	O	O	O					M		
<u>200GBASE-SR2</u>	<u>O</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>M</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>O</u>	<u>O</u>							<u>M</u>
200GBASE-DR4	O	M	O	O	M	M	O	O	O	O	M						
200GBASE-FR4	O	M	O	O	M	M	O	O	O	O		M					
200GBASE-LR4	O	M	O	O	M	M	O	O	O	O			M				
200GBASE-ER4	O	M	O	O	M	M	O	O	O	O				M			

[a] O = Optional, M = Mandatory.

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Change Table 116–5 as follows:

Table 116–5—PHY type and clause correlation (400GBASE optical)

PHY type	Clause[a]																				
	78	117		118	119	120	120B	120C	120D	120E	<u>167</u>		123	138	150	124	122	151		122	
	EEE	RS	400GMII	400GMII Extender	400GBASE-R PCS	400GBASE-R PMA	400GAUI-16 C2C	400GAUI-16 C2M	400GAUI-8 C2C	400GAUI-8 C2M	<u>400GBASE-VR4 PMD</u>	<u>400GBASE-SR4 PMD</u>	400GBASE-SR16 PMD	400GBASE-SR8 PMD	400GBASE-SR4.2	400GBASE-DR4 PMD	400GBASE-FR8 PMD	400GBASE-FR4	400GBASE-LR4-6	400GBASE-LR8 PMD	400GBASE-ER8 PMD
<u>400GBASE-VR4</u>	<u>O</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>M</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>O</u>	<u>O</u>	<u>M</u>										
400GBASE-SR16	O	M	O	O	M	M	O	O	O	O			M								
400GBASE-SR8	O	M	O	O	M	M	O	O	O	O				M							
<u>400GBASE-SR4</u>	<u>O</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>M</u>	<u>M</u>	<u>O</u>	<u>O</u>	<u>O</u>	<u>O</u>		<u>M</u>									
400GBASE-SR4.2	O	M	O	O	M	M	O	O	O	O					M						
400GBASE-DR4	O	M	O	O	M	M	O	O	O	O						M					
400GBASE-FR8	O	M	O	O	M	M	O	O	O	O							M				
400GBASE-FR4	O	M	O	O	M	M	O	O	O	O								M			
400GBASE-LR4-6	O	M	O	O	M	M	O	O	O	O									M		
400GBASE-LR8	O	M	O	O	M	M	O	O	O	O										M	
400GBASE-ER8	O	M	O	O	M	M	O	O	O	O											M

[a] O = Optional, M = Mandatory.

116.2 Summary of 200 Gigabit and 400 Gigabit Ethernet sublayers

116.2.5 Physical Medium Dependent (PMD) sublayer

Change the second paragraph of 116.2.5 as follows:

The 200GBASE-R PMDs and their corresponding media are specified in Clause 121, Clause 122,~~ and~~ Clause 136 through Clause 138<u>, and Clause 167</u>. The 400GBASE-R PMDs and their corresponding media are specified in Clause 122 through Clause 124, Clause 138,~~ and~~ Clause 150, <u>and Clause 167</u>.

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116.4 Delay constraints

Change Table 116–6 as follows (some unchanged rows not shown):

Table 116–6—Sublayer delay constraints (200GBASE)

Sublayer	Maximum (bit time)[a]	Maximum (pause_quanta)[b]	Maximum (ns)	Notes[c]
...				
200GBASE-CR4 PMD	8 192	16	40.96	Includes allocation of 20 ns for one direction through cable medium. See 137.5.
200GBASE-VR2 PMD	4 096	8	20.48	Includes 2 m of fiber. See 167.3.1.
200GBASE-SR4 PMD	4 096	8	20.48	See 138.3.
200GBASE-SR2 PMD	4 096	8	20.48	Includes 2 m of fiber. See 167.3.1.
200GBASE-DR4 PMD	4 096	8	20.48	Includes 2 m of fiber. See 121.3.1.
...				

[a] For 200GBASE-R, 1 bit time (BT) is equal to 5 ps. (See 1.4.215 for the definition of bit time.)
[b] For 200GBASE-R, 1 pause_quantum is equal to 2.56 ns. (See 31B.2 for the definition of pause_quanta.)
[c] Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

Change Table 116–7 as follows (some unchanged rows not shown):

Table 116–7—Sublayer delay constraints (400GBASE)

Sublayer	Maximum (bit time)[a]	Maximum (pause_quanta)[b]	Maximum (ns)	Notes[c]
...				
400GBASE-R PMA	36 864	72	92.16	See 120.5.4.
400GBASE-VR4 PMD	8 192	16	20.48	Includes 2 m of fiber. See 167.3.1.
400GBASE-SR16 PMD	8 192	16	20.48	Includes 2 m of fiber. See 123.3.1.
400GBASE-SR8 PMD	8 192	16	20.48	Includes 2 m of fiber. See 138.3.1.
400GBASE-SR4 PMD	8 192	16	20.48	Includes 2 m of fiber. See 167.3.1.
400GBASE-SR4.2 PMD	8 192	16	20.48	Includes 2 m of fiber. See 150.3.1.
400GBASE-DR4 PMD	8 192	16	20.48	Includes 2 m of fiber. See 124.3.1.
...				

[a] For 400GBASE-R, 1 bit time (BT) is equal to 2.5 ps. (See 1.4.215 for the definition of bit time.)
[b] For 400GBASE-R, 1 pause_quantum is equal to 1.28 ns. (See 31B.2 for the definition of pause_quanta.)
[c] Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

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116.5 Skew constraints

Change Table 116–8 and Table 116–9 as follows:

Table 116–8—Summary of Skew constraints

Skew points	Maximum Skew (ns)[a]	Maximum Skew for 200GBASE-R or 400GBASE-R PCS lane (UI)[b]	Notes[c]
SP1	29	≈ 770	See 120.5.3.1
SP2	43	≈ 1142	See 120.5.3.3, 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP3	54	≈ 1434	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP4	134	≈ 3559	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 138.3.2, 150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP5	145	≈ 3852	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP6	160	≈ 4250	See 120.5.3.5
At PCS receive	180	≈ 4781	See 119.2.5.1

[a] The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.
[b] The symbol ≈ indicates approximate equivalent of maximum Skew in UI based on 1 UI equals 37.64706 ps at PCS lane signaling rate of 26.5625 GBd.
[c] Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

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Table 116–9—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation (ns)	Maximum Skew Variation for 26.5625 GBd PMD lane (UI)[a]	Maximum Skew Variation for 53.125 GBd PMD lane (UI)[b]	Notes[c]
SP1	0.2	≈ 5	N/A	See 120.5.3.1
SP2	0.4	≈ 11	≈ 21	See 120.5.3.3, 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP3	0.6	≈ 16	≈ 32	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP4	3.4	≈ 90	≈ 181	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2,150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP5	3.6	≈ 96	≈ 191	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, ~~or~~ 151.3.2, or 167.3.2
SP6	3.8	≈ 101	N/A	See 120.5.3.5
At PCS receive	4	≈ 106	N/A	See 119.2.5.1

[a] The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 37.64706 ps at PMD lane signaling rate of 26.5625 GBd.
[b] The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.
[c] Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

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Insert Clause 167 after Clause 166 as shown:

167. Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4

167.1 Overview

This clause specifies the 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 PMDs together with the multimode fiber medium. The optical signals generated by these PMD types are modulated using a 4-level pulse amplitude modulation (PAM4) format. The PMD sublayers provide point-to-point 100, 200, and 400 Gigabit Ethernet links over one, two, or four pairs of multimode fiber, respectively.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA, as shown in Table 167–1 or Table 167–2, to the medium through the MDI and optionally with the management functions that may be accessible through the management interface defined in Clause 45, or equivalent. Figure 167–1 shows the relationship of the PMDs and MDIs (shown shaded) with other sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 100 Gigabit Ethernet is introduced in Clause 80 and the purpose of each PHY sublayer is summarized in 80.2. 200 Gigabit Ethernet and 400 Gigabit Ethernet are introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2.

100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 PHYs with the optional Energy-Efficient Ethernet (EEE) fast wake capability may enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization (see Clause 78). The deep sleep mode of EEE is not supported.

The 100GBASE-VR1, 200GBASE-VR2, and 400GBASE-VR4 sublayers provide point-to-point 100, 200, and 400 Gigabit Ethernet links over one, two, or four pairs of multimode fiber, respectively, up to at least 30 m on OM3, 50 m on OM4, and 50 m on OM5 (see 167.7). The 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 sublayers provide point-to-point 100, 200, and 400 Gigabit Ethernet links over one, two, or four pairs of multimode fiber, respectively, up to at least 60 m on OM3, 100 m on OM4, and 100 m on OM5 (see 167.7).

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Table 167–1—Physical Layer clauses associated with the 100GBASE-VR1 and 100GBASE-SR1 PMDs

Associated clause	100GBASE-VR1, 100GBASE-SR1
81—RS	Required
81—CGMII[a]	Optional
82—PCS for 100GBASE-R	Required
83—PMA for 100GBASE-R	Optional
91—RS-FEC[b]	Required
83A—CAUI-10 C2C	Optional
83B—CAUI-10 C2M	Optional
83D—CAUI-4 C2C	Optional
83E—CAUI-4 C2M	Optional
135—PMA for 100GBASE-P	Required
135D—100GAUI-4 C2C	Optional
135E—100GAUI-4 C2M	Optional
135F—100GAUI-2 C2C	Optional
135G—100GAUI-2 C2M	Optional
78—Energy-Efficient Ethernet	Optional

[a]The CGMII is an optional interface. However, if the CGMII is not implemented, a conforming implementation behaves functionally as though the RS and CGMII were present.
[b]The option to bypass the Clause 91 RS-FEC correction function is not supported.

Table 167–2—Physical Layer clauses associated with the 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4 PMDs

Associated clause	200GBASE-VR2, 200GBASE-SR2	400GBASE-VR4, 400GBASE-SR4	Status
117	RS		Required
117	200GMII[a]	400GMII[a]	Optional
118	200GMII Extender	400GMII Extender	Optional
119	PCS for 200GBASE-R	PCS for 400GBASE-R	Required
120	PMA for 200GBASE-R	PMA for 400GBASE-R	Required
120B	200GAUI-8 C2C	400GAUI-16 C2C	Optional
120C	200GAUI-8 C2M	400GAUI-16 C2M	Optional

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Table 167–2—Physical Layer clauses associated with the 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4 PMDs *(continued)*

Associated clause	200GBASE-VR2, 200GBASE-SR2	400GBASE-VR4, 400GBASE-SR4	Status
120D	200GAUI-4 C2C	400GAUI-8 C2C	Optional
120E	200GAUI-4 C2M	400GAUI-8 C2M	Optional
78	Energy-Efficient Ethernet		Optional

[a]The 200GMII and 400GMII are optional interfaces. However, if the appropriate interface is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII or 400GMII were present.

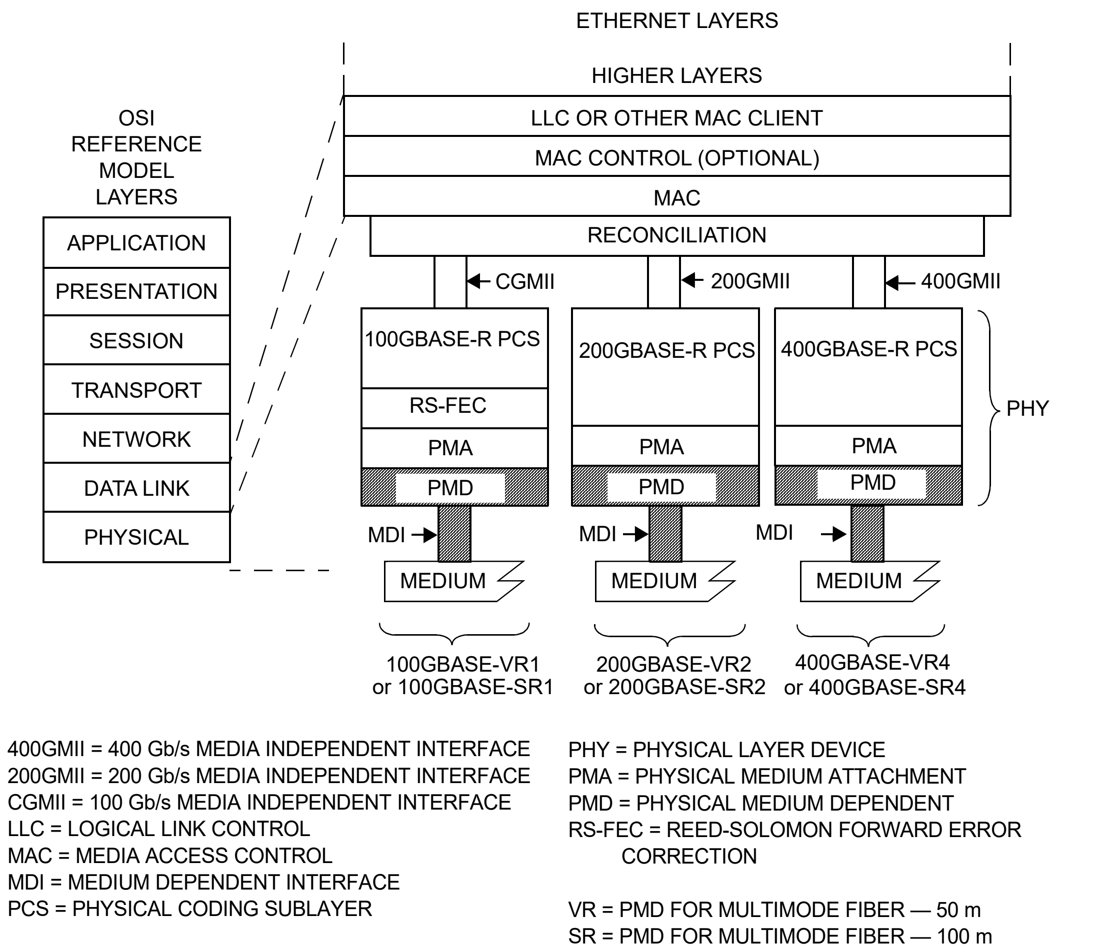

Figure 167–1—100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

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167.1.1 Bit error ratio

For the 100GBASE-VR1 and 100GBASE-SR1 PMDs, the bit error ratio (BER) when processed by the PMA (Clause 135) shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 9.2×10^{-13} for 64-octet frames with minimum interpacket gap when additionally processed by the FEC (Clause 91) and PCS (Clause 82). For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-10} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces. If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 9.2×10^{-13} for 64-octet frames with minimum interpacket gap.

For the 200GBASE-VR2, 200GBASE-SR2, 400GBASE-VR4, and 400GBASE-SR4 PMDs, the bit error ratio (BER) when processed by the PMA (Clause 120) shall be less than 2.4×10^{-4} provided that the error statistics are sufficiently random that this results in a frame loss ratio (see 1.4.344) of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap when additionally processed by the PCS (Clause 119). For a complete Physical Layer, the frame loss ratio may be degraded to 6.2×10^{-11} for 64-octet frames with minimum interpacket gap due to additional errors from the electrical interfaces. If the error statistics are not sufficiently random to meet this requirement, then the BER shall be less than that required to give a frame loss ratio of less than 1.7×10^{-12} for 64-octet frames with minimum interpacket gap.

167.2 Physical Medium Dependent (PMD) service interface

This subclause specifies the services provided by the 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 PMDs. The service interfaces for these PMDs are described in an abstract manner and do not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The 100GBASE-VR1 or 100GBASE-SR1 PMD service interface is an instance of the inter-sublayer service interface defined in 80.3, with a single symbol stream (n = 1).

The 200GBASE-VR2 or 200GBASE-SR2 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with two parallel symbol streams (n = 2).

The 400GBASE-VR4 or 400GBASE-SR4 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with four parallel symbol streams (n = 4).

The service interface primitives are summarized as follows:

PMD:IS_UNITDATA_*i*.request
PMD:IS_UNITDATA_*i*.indication
PMD:IS_SIGNAL.indication

The 100GBASE-VR1 or 100GBASE-SR1 PMD has a single symbol stream, hence $i = 0$.

The 200GBASE-VR2 or 200GBASE-SR2 PMD has two parallel symbol streams, hence $i = 0$ to 1.

The 400GBASE-VR4 or 400GBASE-SR4 PMD has four parallel symbol streams, hence $i = 0$ to 3.

In the transmit direction, the PMA continuously sends n streams of PAM4 symbols to the PMD, one per lane, using the PMD:IS_UNITDATA_*i*.request primitive, at a nominal signaling rate of 53.125 GBd. The PMD converts these streams of symbols into appropriate signals on the MDI.

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In the receive direction, the PMD continuously sends *n* streams of PAM4 symbols to the PMA, corresponding to the signals received from the MDI, one per lane, using the PMD:IS_UNITDATA_*i*.indication primitive, at a nominal signaling rate of 53.125 GBd.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable SIGNAL_DETECT parameter as defined in 167.5.4. The SIGNAL_DETECT parameter can take on one of two values: OK or FAIL. When SIGNAL_DETECT = FAIL, the rx_symbol parameters are undefined.

NOTE—SIGNAL_DETECT = OK does not guarantee that the rx_symbol parameters are known to be good. It is possible for a poor quality link to provide sufficient light for a SIGNAL_DETECT = OK indication and still not meet the BER defined in 167.1.1.

167.3 Delay and Skew

167.3.1 Delay constraints

An upper bound to the delay through the PMA and PMD is required for predictable operation of the MAC Control PAUSE operation.

The sum of the transmit and receive delays at one end of the link contributed by the 100GBASE-VR1 or 100GBASE-SR1 PMD including 2 m of fiber in one direction shall be no more than 2048 bit times (4 pause_quanta or 20.48 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 200GBASE-VR2 or 200GBASE-SR2 PMD including 2 m of fiber in one direction shall be no more than 4096 bit times (8 pause_quanta or 20.48 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-VR4 or 400GBASE-SR4 PMD including 2 m of fiber in one direction shall be no more than 8192 bit times (16 pause_quanta or 20.48 ns).

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 80.4 for 100GBASE-VR1 and 100GBASE-SR1, and in 116.4 and its references for 200GBASE-VR2, 200GBASE-SR2, 400GBASE-VR4, and 400GBASE-SR4.

167.3.2 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation also needs to be limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

Skew and Skew Variation are defined in 80.5 and 116.5 and specified at the points SP1 to SP6 shown in Figure 80–8 and Figure 116–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps. For 100GBASE-VR1 and 100GBASE-SR1, since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

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If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns. For 100GBASE-VR1 and 100GBASE-SR1, since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation see 80.5 and 116.5.

167.4 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several variables that may provide control and status information for and about the PMD. If the MDIO interface is implemented, the mapping of MDIO control variables to PMD control variables shall be as shown in Table 167–3, and the mapping of MDIO status variables to PMD status variables shall be as shown in Table 167–4.

Table 167–3—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1 register	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable register	1.9.0	PMD_global_transmit_disable
PMD transmit disable *n*–1 to PMD transmit disable 0	PMD transmit disable register	1.9.*n* to 1.9.1	PMD_transmit_disable_*n*–1 to PMD_transmit_disable_0

Table 167–4—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/ bit number	PMD status variable
Fault	PMA/PMD status 1 register	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2 register	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2 register	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect register	1.10.0	PMD_global_signal_detect
PMD receive signal detect *n*–1 to PMD receive signal detect 0	PMD receive signal detect register	1.10.*n* to 1.10.1	PMD_signal_detect_*n*–1 to PMD_signal_detect_0

167.5 PMD functional specifications

The 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

167.5.1 PMD block diagram

The PMD block diagram for 400GBASE-VR4 or 400GBASE-SR4 is shown in Figure 167–2. The block diagrams for 200GBASE-VR2 and 200GBASE-SR2 are equivalent to Figure 167–2, but for two lanes per

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direction. The block diagrams for 100GBASE-VR1 and 100GBASE-SR1 are equivalent to Figure 167–2, but for one lane per direction.

For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a multimode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 167.8 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 167.10.3). Unless specified otherwise, all receiver measurements and tests defined in 167.8 are made at TP3.

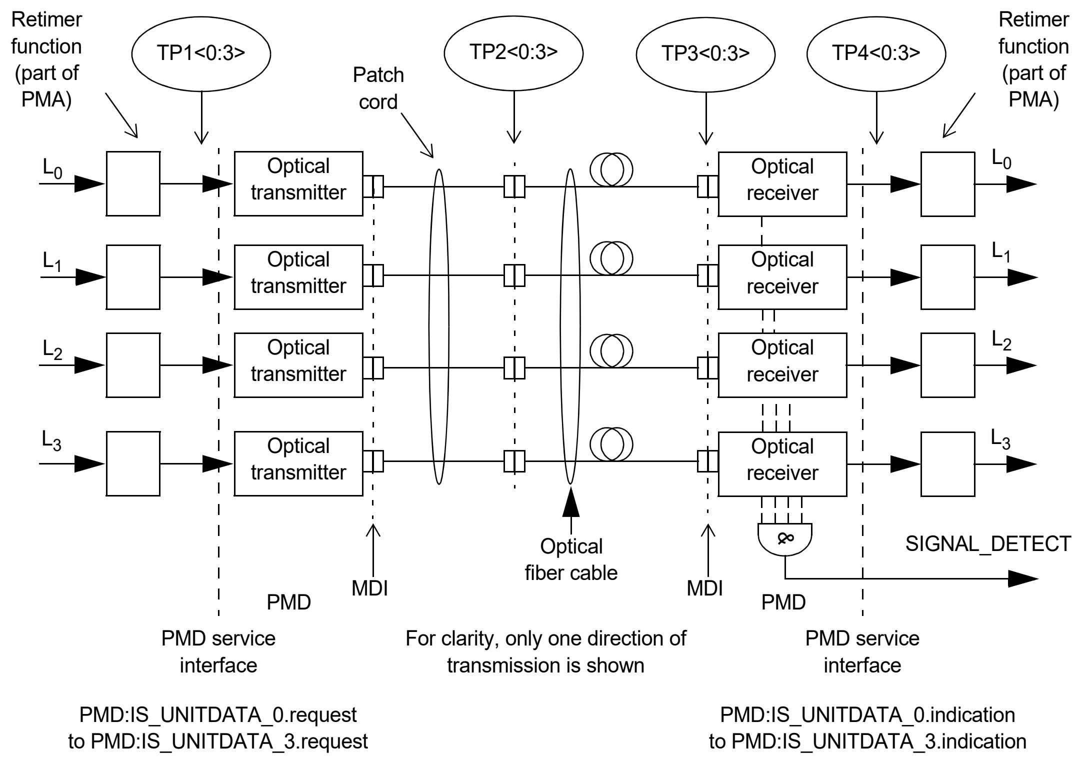

Figure 167–2—Block diagram for 400GBASE-VR4 or 400GBASE-SR4 transmit/receive paths

TP1<0:3> and TP4<0:3> are optional reference points that may be useful to implementers for testing components (these test points might not be accessible in an implemented system).

167.5.2 PMD transmit function

The PMD Transmit function shall convert the one, two, or four symbol streams requested by the PMD service interface messages PMD:IS_UNITDATA_*i*.request into one, two, or four separate optical signals. The 100GBASE-VR1 and 100GBASE-SR1 PMDs have a single symbol stream, hence i = 0. The 200GBASE-VR2 and 200GBASE-SR2 PMDs have two parallel symbol streams, hence i = 0 to 1. The 400GBASE-VR4 and 400GBASE-SR4 PMDs have four parallel symbol streams, hence i = 0 to 3. Each optical signal shall then be delivered to the MDI, which contains one, two, or four parallel light paths for transmit, according to the transmit optical specifications in this clause. The four optical power levels in each signal in order from lowest to highest shall correspond to tx_symbol values zero, one, two, and three, respectively.

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167.5.3 PMD receive function

The PMD Receive function shall convert the one, two, or four parallel optical signals received from the MDI into one, two, or four separate symbol streams for delivery to the PMD service interface using the messages PMD:IS_UNITDATA_0.indication to PMD:IS_UNITDATA_3.indication, all according to the receive optical specifications in this clause. The four optical power levels in each signal in order from lowest to highest shall correspond to rx_symbol values zero, one, two, and three, respectively.

167.5.4 PMD global signal detect function

The PMD global signal detect function shall report the state of SIGNAL_DETECT via the PMD service interface. The SIGNAL_DETECT parameter is signaled continuously, while the PMD:IS_SIGNAL.indication message is generated when a change in the value of SIGNAL_DETECT occurs. The SIGNAL_DETECT parameter defined in this clause maps to the SIGNAL_OK parameter in the inter-sublayer service interface primitives defined in 167.2.

SIGNAL_DETECT shall be a global indicator of the presence of optical signals on all lanes. The value of the SIGNAL_DETECT parameter shall be generated according to the conditions defined in Table 167–5. The PMD receiver is not required to verify whether a compliant 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 signal is being received. This standard imposes no response time requirements on the generation of the SIGNAL_DETECT parameter.

Table 167–5—SIGNAL_DETECT value definition

Receive conditions	SIGNAL_DETECT value
For any lane; Average optical power at TP3 ≤ –30 dBm	FAIL
For all lanes; [(Optical power at TP3 ≥ average receive power, each lane (min) in Table 167–8) AND (compliant 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 signal input)]	OK
All other conditions	Unspecified

As a consequence of the requirements for the setting of the SIGNAL_DETECT parameter, implementations need to provide adequate margin between the input optical power level at which the SIGNAL_DETECT parameter is set to OK, and the inherent noise level of the PMD including the effects of crosstalk, power supply noise, etc.

Various implementations of the Signal Detect function are permitted by this standard, including implementations that generate the SIGNAL_DETECT parameter values in response to the amplitude of the modulation of the optical signal and implementations that respond to the average optical power of the modulated optical signal.

167.5.5 PMD lane-by-lane signal detect function

Various implementations of the Signal Detect function are permitted by this standard. When the MDIO is implemented, each PMD_signal_detect_*i*, where *i* represents the lane number in the range 0:3, shall be continuously set in response to the magnitude of the optical signal on its associated lane, according to the requirements of Table 167–5.

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167.5.6 PMD reset function

If the MDIO interface is implemented, and if PMD_reset is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

167.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

a) When the PMD_global_transmit_disable variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 167–7.
b) If a PMD_fault is detected, then the PMD may set the PMD_global_transmit_disable variable to one, turning off the optical transmitter in each lane.

167.5.8 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the optical transmitter in each lane to be selectively disabled.

a) When a PMD_transmit_disable_*i* variable (where *i* represents the lane number in the range 0:3) is set to one, this function shall turn off the optical transmitter associated with that variable so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 167–7.
b) If a PMD_fault is detected, then the PMD may set each PMD_transmit_disable_*i* to one, turning off the optical transmitter in each lane.

If the optional PMD lane-by-lane transmit disable function is not implemented in MDIO, an alternative method may be provided to independently disable each transmit lane.

167.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault to one.

If the MDIO interface is implemented, PMD_fault shall be mapped to the fault bit as specified in 45.2.1.2.3.

167.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set PMD_transmit_fault to one.

If the MDIO interface is implemented, PMD_transmit_fault shall be mapped to the transmit fault bit as specified in 45.2.1.7.4.

167.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

If the MDIO interface is implemented, PMD_receive_fault shall be mapped to the receive fault bit as specified in 45.2.1.7.5.

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167.6 Lane assignments

There are no lane assignments (within a group of transmit or receive lanes) for 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, or 400GBASE-SR4. While it is expected that a PMD will map electrical lane *i* to optical lane *i* and vice versa, there is no need to define the physical ordering of the lanes, as the PCS sublayer is capable of receiving the lanes in any arrangement. The positioning of transmit and receive lanes at the MDI is specified in 167.10.3.1.

167.7 PMD to MDI optical specifications for 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4

The operating ranges for the 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 PMDs are defined in Table 167–6. A compliant PMD operates on 50/125 µm multimode fibers, type A1a.2 (OM3), type A1a.3 (OM4), or type A1a.4 (OM5), according to the specifications defined in Table 167–14. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 100GBASE-SR1 PMD operating at 120 m meets the operating range requirement of 0.5 m to 100 m).

Table 167–6—Operating range

PMD type	Required operating range[a]
100GBASE-VR1 200GBASE-VR2 400GBASE-VR4	0.5 m to 30 m for OM3
	0.5 m to 50 m for OM4
	0.5 m to 50 m for OM5
100GBASE-SR1 200GBASE-SR2 400GBASE-SR4	0.5 m to 60 m for OM3
	0.5 m to 100 m for OM4
	0.5 m to 100 m for OM5

[a]The RS-FEC correction function may not be bypassed for any operating distance.

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167.7.1 Transmitter optical specifications

Each lane of a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 transmitter shall meet the specifications in Table 167–7 per the definitions in 167.8.

Table 167–7—Transmit characteristics

Description	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4	100GBASE-SR1 200GBASE-SR2 400GBASE-SR4	Unit
Signaling rate, each lane (range)	53.125 ± 100 ppm		GBd
Modulation format	PAM4		—
Center wavelength (range)	842 to 948	844 to 863	nm
RMS spectral width[a] (max)	0.65	0.6	nm
Average launch power, each lane (max)	4		dBm
Average launch power, each lane (min)	–4.6		dBm
Outer Optical Modulation Amplitude (OMA_{outer}), each lane (max)	3.5		dBm
Outer Optical Modulation Amplitude (OMA_{outer}), each lane (min) For max (TECQ, TDECQ) ≤ 1.8 dB For 1.8 < max (TECQ, TDECQ) ≤ 4.4 dB	 –2.6 –4.4 + max (TECQ, TDECQ)		 dBm dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	4.4	4.4	dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	4.4		dB
Overshoot/undershoot (max)	29		%
Transmitter power excursion, each lane (max)	2.3		dBm
Extinction ratio, each lane (min)	2.5		dB
Transmitter transition time, each lane (max)	17		ps
Average launch power of OFF transmitter, each lane (max)	–30		dBm
$RIN_{14}OMA$ (max)	–132		dB/Hz
Optical return loss tolerance (max)	14		dB
Encircled flux[b]	≥ 86% at 19 µm ≤ 30% at 4.5 µm		—

[a]RMS spectral width is the standard deviation of the spectrum.
[b]If measured into type A1a.2 or type A1a.3, or A1a.4, 50 µm fiber, in accordance with IEC 61280-1-4.

The values of OMA_{outer}, each lane (max) and OMA_{outer}, each lane (min) and their dependence on TDECQ and TECQ are illustrated in Figure 167.8.

167.7.2 Receiver optical specifications

Each lane of a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 receiver shall meet the specifications in Table 167–8 per the definitions in 167.8.

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Table 167–8—Receive characteristics

Description	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4	100GBASE-SR1 200GBASE-SR2 400GBASE-SR4	Unit
Signaling rate, each lane (range)	53.125 ± 100 ppm		GBd
Modulation format	PAM4		—
Center wavelength (range)	842 to 948		nm
Damage threshold[a] (min)	5		dBm
Average receive power, each lane (max)	4		dBm
Average receive power, each lane[b] (min)	–6.3	–6.4	dBm
Receive power, each lane (OMA_{outer}) (max)	3.5		dBm
Receiver reflectance (max)	–15		dB
Receiver sensitivity (OMA_{outer}) (max) For TECQ ≤ 1.8 dB For 1.8 < TECQ ≤ 4.4 dB	 –4.4 –6.2 + TECQ	 –4.6 –6.4 + TECQ	 dBm dBm
Stressed receiver sensitivity (OMA_{outer})[c] (max)	–1.8	–2.0	dBm
Conditions of stressed receiver sensitivity test:[d]			
Stressed eye closure for PAM4 (SECQ), lane under test	4.4		dB
OMA_{outer} of each aggressor lane	3.5		dBm

[a]The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level on one lane. The receiver does not have to operate correctly at this input power.
[b]Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.
[c]Measured with conformance test signal at TP3 (see 167.8.14) for the BER specified in 167.1.1.
[d]These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

The value of receiver sensitivity (OMA_{outer}) (max) in Table 167–8 depends on TECQ of the test source as illustrated in Figure 167.8.

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167.7.3 Illustrative link power budget

An illustrative power budget and penalties for 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 channels are shown in Table 167–9.

Table 167–9—Illustrative link power budget

Parameter	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4			100GBASE-SR1 200GBASE-SR2 400GBASE-SR4			Unit
	OM3	OM4	OM5	OM3	OM4	OM5	
Effective modal bandwidth at 850 nm[a]	2000	4700		2000	4700		MHz · km
Power budget (for max TDECQ)	6.2			6.4			dB
Operating distance	0.5 to 30	0.5 to 50		0.5 to 60	0.5 to 100		m
Channel insertion loss[b]	1.6	1.7		1.7	1.8		dB
Allocation for penalties[c] (for max TDECQ)	4.5			4.6			dB
Additional insertion loss allowed	0.1	0		0.1	0		dB

[a]Per IEC 60793-2-10, see Table 167–15 for information on effective modal bandwidth at other wavelengths.
[b]The channel insertion loss is calculated using the maximum distance specified in Table 167–6 and cabled optical fiber attenuation of 3.0 dB/km at 850 nm plus an allocation for connection and splice loss given in 167.10.2.2.1.
[c]Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested.

The values of transmitter OMA_{outer}(max), transmittter OMA_{outer}(min) versus TDECQ and TECQ, and the receiver sensitivity (OMA_{outer})(max) versus TECQ are illustrated in Figure 167–3.

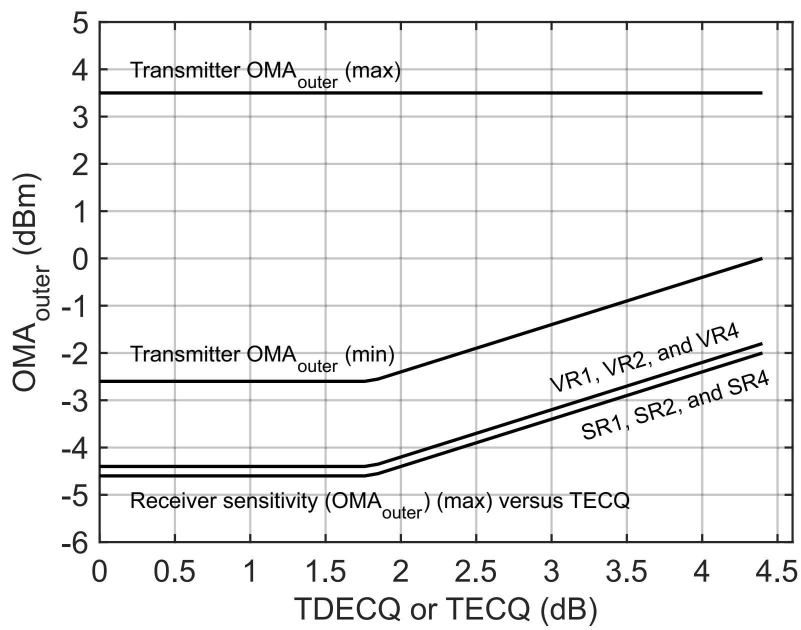

Figure 167–3—Transmitter OMA_{outer}, each lane versus TDECQ and TECQ, and receiver sensitivity (OMA_{outer}) versus TECQ

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167.8 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

167.8.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 167–11 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 167–11 may be used to perform that test. The test patterns used in this clause are shown in Table 167–10.

Table 167–10—Test patterns

Pattern	Pattern description	Defined in
Square wave	Square wave (8 threes, 8 zeros)	120.5.11.2.4
3	PRBS31Q	120.5.11.2.2
4	PRBS13Q	120.5.11.2.1
5	Scrambled idle encoded by RS-FEC	82.2.11 and 91, or 119.2.4.9
6	SSPRQ	120.5.11.2.3

Table 167–11—Test-pattern definitions and related subclauses

Parameter	Pattern	Related subclause
Wavelength, spectral width	3, 4, 5, 6, or valid 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 signal	167.8.3
Average optical power	3, 4, 5, 6, or valid 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 signal	167.8.4
Outer Optical Modulation Amplitude (OMA_{outer})	4 or 6	167.8.5
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	167.8.6
Transmitter eye closure for PAM4 (TECQ)	6	167.8.7
Overshoot/undershoot	6	167.8.8
Transmitter power excursion	6	167.8.9
Extinction ratio	4 or 6	167.8.10
Transmitter transition time	Square wave or 6	167.8.11
$RIN_{14}OMA$	Square wave	167.8.12
Receiver sensitivity	3 or 5	167.8.13

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Table 167–11—Test-pattern definitions and related subclauses *(continued)*

Parameter	Pattern	Related subclause
Stressed receiver sensitivity	3, 5, or valid 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, 400GBASE-SR4 signal	167.8.14
Stressed eye closure (SEC), calibration	6	167.8.14

167.8.2 Multi-lane testing considerations

Receiver sensitivity and stressed receiver sensitivity are defined for an interface at the BER specified in 167.1.1. The interface BER is the average of the BERs of the receive lanes when they are stressed. Measurements with Pattern 3 (PRBS31Q) allow lane-by-lane BER measurements. Measurement with Pattern 5 (scrambled idle encoded by RS-FEC) gives the interface BER if all lanes are stressed at the same time.

If each lane is stressed in turn, the BER is diluted by the unstressed lanes, and the BER for that stressed lane alone is found, e.g., by multiplying by four for 400GBASE-SR4 if the unstressed lanes have low BER. In stressed receiver sensitivity measurements, unstressed lanes may be created by setting the power at the receiver under test well above its sensitivity and/or not stressing those lanes with ISI and jitter, or by other means. Each receive lane is stressed in turn while all are operated. All aggressor lanes are operated as specified. To find the interface BER, the BERs of all lanes when stressed are averaged.

Where relevant, parameters are defined with all co-propagating and counter-propagating lanes operational so that crosstalk effects are included. Where not otherwise specified, the maximum amplitude (OMA_{outer}) for a particular situation is used. While the lanes in a particular direction may share a common clock, the Tx and Rx directions are not synchronous to each other. If Pattern 3 is used for the lanes not under test using a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane so that the symbols on each lane are not correlated within the PMD. Alternative test methods that generate equivalent results may be used.

167.8.3 Center wavelength and spectral width

The center wavelength and RMS spectral width of each optical lane shall be within the range given in Table 167–7 if measured per IEC 61280-1-3. The lane under test is modulated using one of the test patterns specified in Table 167–11.

167.8.4 Average optical power

The average optical power of each lane shall be within the limits given in Table 167–7 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using one of the test patterns defined in Table 167–11.

167.8.5 Outer Optical Modulation Amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 167–7 if measured as defined in 121.8.4.

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167.8.6 Transmitter and dispersion eye closure for PAM4 (TDECQ)

TDECQ is a measure of each optical transmitter's vertical eye closure as measured through an optical to electrical converter (O/E) with a bandwidth equivalent to a combined reference receiver and worst case optical channel, and equalized with the reference equalizer specified in 167.8.6.1. Table 167–11 specifies the test pattern to be used for measurement of TDECQ.

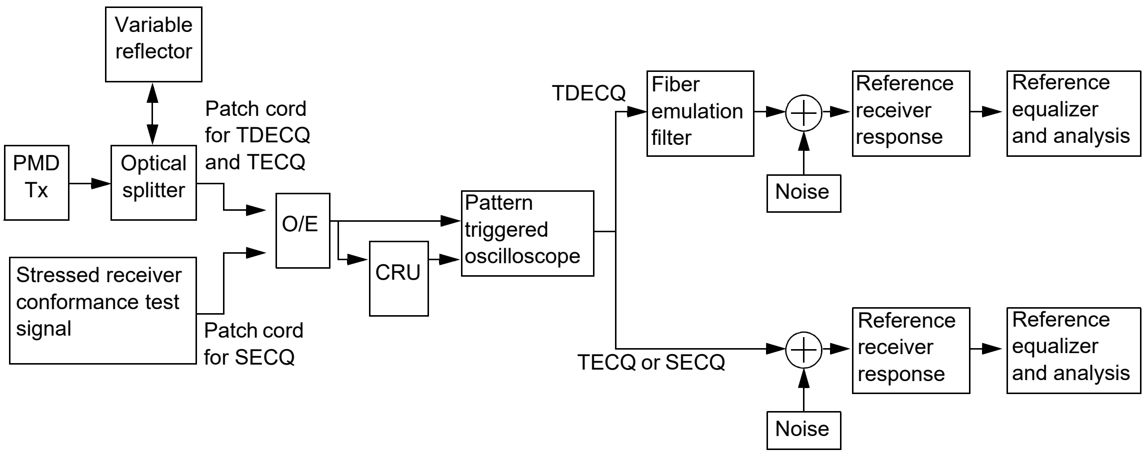

Figure 167–4—TDECQ, TECQ, and SECQ conformance test block diagram

The TDECQ of each lane shall be within the limits given in Table 167–7 if measured using the methods specified in 121.8.5, with the following exceptions:

- The conformance test block diagram shown in Figure 167–4 is used in place of Figure 121–4.
- The optical channel requirements in 121.8.5.2 do not apply. Instead, the optical splitter and variable reflector are adjusted so that each transmitter is tested with an optical return loss equal to the "optical return loss tolerance (max)" given in Table 167–7.
- The frequency response of the combination of the O/E converter and the oscilloscope used to measure the optical waveform is that of two cascaded filters. The first filter emulates the dispersion of the fiber and has a 3 dB bandwidth f_A that depends on the PMD type and the center wavelength of the transmitter according to Table 167–12. The fourth-order Bessel-Thomson response of the first filter should extend to at least $2.6 \times f_A$ GHz, and at frequencies above $2.6 \times f_A$ GHz, the response should not exceed –20 dB. The second filter represents the reference equalizer front end and has a 3 dB bandwidth of approximately 26.5625 GHz with a fourth-order Bessel-Thomson response to at least 1.3 × 53.125 GHz. At frequencies above 1.3 × 53.125 GHz, the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.
- The normalized noise power density spectrum, N(f) in Equation (121-9), is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a bandwidth of 26.5625 GHz.
- If an equivalent-time sampling oscilloscope is used, the impact of the sampling process and the fiber emulation is compensated for, so that the correct magnitude of noise is present at the output of the equalizer.
- P_{th1}, P_{th2}, and P_{th3} are varied from their nominal values by up to ±2% of OMA_{outer} in order to optimize TDECQ. The same three thresholds are used for both the left and the right histogram.
- The reference equalizer to be used for TDECQ is specified in 167.8.6.1.

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Table 167–12—The 3 dB bandwidth f_A of the fiber emulation filter for TDECQ measurement

PMD type	Center wavelength (range) (nm)	3 dB bandwidth f_A (GHz)
100GBASE-VR1 200GBASE-VR2 400GBASE-VR4	842 to 868	33.6
	842 to 888	29.6
	842 to 918	24.5
	842 to 948	20.7
100GBASE-SR1 200GBASE-SR2 400GBASE-SR4	844 to 863	18.0

The lowest measured TDECQ values are achieved with the equalizer optimization method described in 121.8.5. Alternative optimization methods such as minimum mean squared error (MMSE) may be used to determine equalizer tap weights to reduce test time, and are expected to report equal or higher values of TDECQ. These alternative methods should not be used for receiver sensitivity and stressed receiver sensitivity calibration.

167.8.6.1 TDECQ reference equalizer

A functional model of the 9-tap reference equalizer is shown in Figure 167–5. The sum of the equalizer tap coefficients is equal to 1. Tap 2, tap 3, or tap 4 has the largest magnitude tap coefficient, which is constrained to be at least 0.8. The absolute values of tap 7, tap 8, and tap 9 are constrained to be less than 0.3, 0.2, and 0.1, respectively.

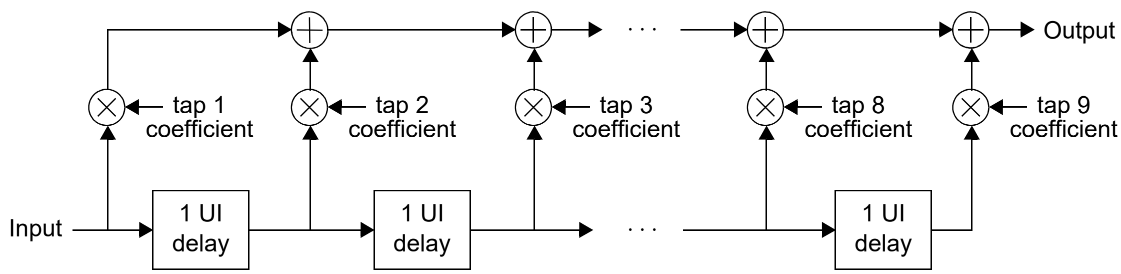

Figure 167–5—TDECQ and TECQ reference equalizer functional model

NOTE—This reference equalizer is part of the TDECQ and TECQ tests and does not imply any particular receiver implementation.

167.8.7 Transmitter eye closure for PAM4 (TECQ)

The transmitter eye closure for PAM4 (TECQ) is a measure of the optical transmitter's eye closure at TP2. A block diagram for the TECQ conformance test is shown in Figure 167–4. The TECQ of each lane shall be within the limits given in Table 167–7 if measured using a test pattern specified for TECQ in Table 167–11. The TECQ of each lane is measured using the methods specified for TDECQ in 167.8.6 except that the first filter representing the dispersion of the fiber is omitted. The reference equalizer to be used for TECQ is specified in 167.8.6.1.

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167.8.8 Overshoot/undershoot

The overshoot/undershoot of each lane shall be within the limits given in Table 167–7 if measured using a test pattern specified for overshoot/undershoot in Table 167–11.

Overshoot and undershoot are measured using the waveform captured for the TECQ test (see 167.8.7), but without the reference equalizer being applied.

Overshoot and undershoot are calculated relative to OMA_{outer} using the methods in 140.7.7 except that a hit ratio of 3×10^{-3} is used.

167.8.9 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limits given in Table 167–7 if measured using a test pattern specified for transmitter power excursion in Table 167–11.

Transmitter power excursion is measured using the waveforms captured for the TECQ test (see 167.8.7), but without the reference equalizer being applied. Transmitter power excursion is defined as:

Transmitter power excursion = max ($P_{max} - P_{average}$, $P_{average} - P_{min}$)

where

P_{max} and P_{min}	are defined by the methods in 140.7.7 using a 3×10^{-3} hit ratio
$P_{average}$	is the average optical power in the lane under test defined in 167.8.4

167.8.10 Extinction ratio

The extinction ratio of each lane shall be within the limits given in Table 167–7 if measured using the methods specified in 121.8.6.

167.8.11 Transmitter transition time

The transmitter transition time of each lane shall be within the limits given in Table 167–7 if measured using a test pattern specified for transmitter transition time in Table 167–11.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer}, or from 80% of OMA_{outer} to 20% of OMA_{outer}, for the rising and falling edges respectively, as measured through an O/E converter and oscilloscope with a combined 3 dB bandwidth of approximately 26.5625 GHz with a fourth-order Bessel-Thomson response to at least 1.3 × 53.125 GHz. At frequencies above 1.3 × 53.125 GHz the response should not exceed –20 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 167.8.5), with the exception that the square wave test pattern can be used. When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence and the falling edge is that within the 33333000000 symbol sequence.

167.8.12 Relative intensity noise ($RIN_{14}OMA$)

RIN shall be as defined by the measurement methodology of 52.9.6 with the following exceptions:

a) The optical return loss is 14 dB.
b) The upper –3 dB limit of the measurement apparatus is approximately equal to 26.5625 GHz.
c) The test pattern is according to Table 167–11.

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167.8.13 Receiver sensitivity

The receiver sensitivity (OMA_{outer}) shall be within the limits given in Table 167–8 if measured using a test pattern specified for receiver sensitivity in Table 167–11. Receiver sensitivity (OMA_{outer}) (max) depends on TECQ and is illustrated in Figure 167–3.

The conformance test signal at TP3 meets the requirements for a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 transmitter followed by an attenuator. For multi-lane testing considerations, see 167.8.2. The BER requirements are given in 167.1.1.

The TECQ of the conformance test signal is measured according to 167.8.7. The measured value of TECQ is then used to calculate the limit for receiver sensitivity (OMA_{outer}) as specified in Table 167–8.

167.8.14 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limits given in Table 167–8 if measured using the methodology defined in 121.8.10.1 and 121.8.10.3, with the conformance test signal at TP3 as described in 121.8.10.2, with the following exceptions:

- The stressed receiver conformance test signal has a transition time that is no greater than the value specified in Table 167–7.
- The SECQ of the stressed receiver conformance test signal is equal to the value of TECQ of the signal measured according to 167.8.7. The optical splitter and variable reflector are omitted as shown in Figure 167–4.
- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the $RIN_{14}OMA$ of the SRS test source should be no greater than the $RIN_{14}OMA$ (max) specified for the transmit characteristics in Table 167–7.
- The test patterns used for stressed receiver sensitivity are specified in Table 167–11.
- The applied sinusoidal jitter is specified in 167.8.14.1.
- The values of overshoot/undershoot and transmitter power excursion of the stressed receiver conformance signal are within the limits specified in Table 167–7.
- For a receiver in a multilane device, the OMA_{outer} of the aggressor lanes is specified in Table 167–8.

The relevant BER is the interface BER; see 167.1.1 and 167.8.2. Stressed receiver sensitivity is defined with all transmit and receive lanes in operation. Any of the patterns specified for stressed receiver sensitivity in Table 167–11 is sent from the transmit section of the PMD under test. The signal being transmitted is asynchronous to the received signal.

167.8.14.1 Sinusoidal jitter for receiver conformance test

The sinusoidal jitter is used to test receiver jitter tolerance. The amplitude of the applied sinusoidal jitter is dependent on frequency as specified in Table 167–13.

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Table 167–13—Applied sinusoidal jitter

Frequency range	Sinusoidal jitter peak-to-peak (UI)
$f < 40$ kHz	Not specified
40 kHz $< f \leq$ 4 MHz	2×10^5 Hz/f
4 MHz $< f \leq$ 10 LB[a]	0.05

[a]LB = loop bandwidth; upper frequency bound for added sinusoidal jitter should be at least 10 times the loop bandwidth of the receiver being tested.

167.9 Safety, installation, environment, and labeling

167.9.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

167.9.2 Laser safety

100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 optical transceivers shall conform to Hazard Level 1M laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.[8]

167.9.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

167.9.4 Environment

Normative specifications in this clause shall be met by a system integrating a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate, in the literature associated with the PHY, the operating environmental conditions to facilitate selection, installation, and maintenance.

[8]A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

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It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

167.9.5 Electromagnetic emission

A system integrating a 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

167.9.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

167.9.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4). It is recommended that each PHY with an angled fiber interface indicate that it uses an angled MDI.

Labeling requirements for Hazard Level 1M lasers are given in the laser safety standards referenced in 167.9.2.

167.10 Fiber optic cabling model

The fiber optic cabling (channel) contains one optical fiber for each direction to support 100GBASE-VR1 or 100GBASE-SR1, two optical fibers for each direction to support 200GBASE-VR2 or 200GBASE-SR2, or four optical fibers for each direction to support 400GBASE-VR4 or 400GBASE-SR4. The fiber optic cabling interconnects the transmitters at the MDI on one end of the channel to the receivers at the MDI on the other end of the channel. As defined in 167.10.3, the optical lanes appear in defined locations at the MDI but the locations are not assigned specific lane numbers within this standard because any transmitter lane may be connected to any receiver lane.

167.10.1 Fiber optic cabling model

The fiber optic cabling model is shown in Figure 167–6.

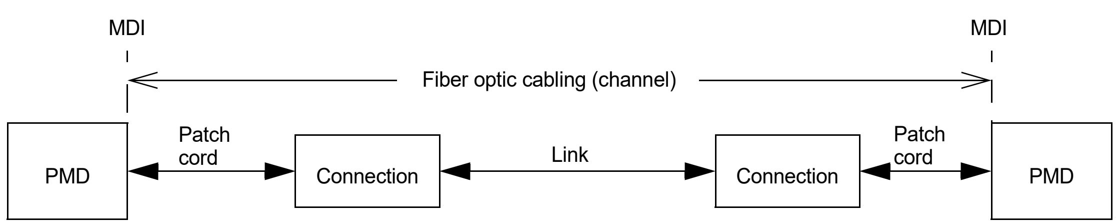

Figure 167–6—Fiber optic cabling model

The channel insertion loss is given in Table 167–14. A channel may contain additional connectors as long as the optical characteristics of the channel (such as attenuation, modal dispersion, reflections and losses of all connectors and splices) meet the specifications. Insertion loss measurements of installed fiber cables are

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made in accordance with IEC 61280-4-1:2009. As OM4 and OM5 optical fiber meet the requirements for OM3, a channel compliant to the “OM3” column may use OM4 or OM5 optical fiber, or a combination of OM3, OM4, and OM5. The fiber optic cabling model (channel) defined here is the same as a simplex fiber optic link segment. The term *channel* is used here for consistency with generic cabling standards.

Table 167–14—Fiber optic cabling (channel) characteristics

Description	100GBASE-VR1 200GBASE-VR2 400GBASE-VR4			100GBASE-SR1 200GBASE-SR2 400GBASE-SR4			Unit
	OM3	OM4	OM5	OM3	OM4	OM5	
Operating distance (max)	30	50		60	100		m
Cabling Skew (max)[a]	79						ns
Cabling Skew Variation[a, b] (max)	2.4						ns
Channel insertion loss[c] (max)	1.6	1.7		1.7	1.8		dB
Channel insertion loss (min)	0			0			dB

[a]Applies only to 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4.
[b]An additional 400 ps of Skew Variation could be caused by wavelength changes, which are attributable to the transmitter not the channel.
[c]These channel insertion loss values include cable loss plus 1.5 dB allocated for connection and splice loss, over the wavelength range 842 nm to 948 nm for VR and over the wavelength range 844 nm to 863 nm for SR.

167.10.2 Characteristics of the fiber optic cabling (channel)

The fiber optic cabling shall meet the specifications defined in Table 167–14. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together.

167.10.2.1 Optical fiber cable

The fiber contained within the fiber optic cabling shall comply with the specifications and parameters of Table 167–15. A variety of multimode cable types may satisfy these requirements, provided the resulting channel also meets the specifications of Table 167–14.

167.10.2.2 Optical fiber connection

An optical fiber connection, as shown in Figure 167–6, consists of a mated pair of optical connectors.

167.10.2.2.1 Connection insertion loss

The maximum link distance is based on an allocation of 1.5 dB total connection and splice loss. For example, this allocation supports three connections with an average insertion loss per connection of 0.5 dB. Connections with lower loss characteristics may be used provided the requirements of Table 167–14 are met. However, the loss of a single connection shall not exceed 0.75 dB.

167.10.2.2.2 Maximum discrete reflectance

The maximum discrete reflectance shall be less than –20 dB.

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Table 167–15—Optical fiber and cable characteristics

Description	OM3[a]	OM4[b]	OM5[c]	Unit
Nominal core diameter	50			μm
Nominal fiber specification wavelength	850		850 and 953	nm
Effective modal bandwidth at 850 nm (min)[d]	2000	4700		MHz · km
Effective modal bandwidth at 953 nm (min)	Not Specified[e]		2470[e]	MHz · km
Cabled optical fiber attenuation (max)	3.0			dB/km
Zero dispersion wavelength (λ_0)[f]	$1297 \le \lambda_0 \le 1328$			nm
Chromatic dispersion slope (max) (S_0)[f]	$-412/(840(1-(\lambda_0/840)^4))$			ps/nm^2 km

[a]IEC 60793-2-10 type A1a.2.
[b]IEC 60793-2-10 type A1a.3.
[c]IEC 60793-2-10 type A1a.4.
[d]When measured with the launch conditions specified in Table 167–7.
[e]Guidance is provided for effective modal bandwidth at all wavelengths in the 840 nm to 953 nm range in IEC 60793-2-10.
[f]These limits are consistent with IEC 60793-2-10/AMD1 ED7. For OM5, they are the same as previous versions of IEC 60793-2-10. OM3 and OM4 fibers compliant to previous versions of IEC 60793-2-10 are considered compliant for 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4 Physical Layer types.

167.10.3 Medium Dependent Interface (MDI)

The 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, or 400GBASE-SR4 PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the “fiber optic cabling” (as shown in Figure 167–6). Examples of an MDI include the following:

a) PMD with a connectorized fiber pigtail plugged into an adapter;
b) PMD receptacle.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 167.5.1, not at the MDI.

167.10.3.1 Optical lane assignments for 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4

The two transmit and two receive optical lanes of 200GBASE-VR2 or 200GBASE-SR2 shall occupy the positions depicted in Figure 167–7 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains four active lanes within 12 total positions. The eight center positions are unused. The transmit optical lanes occupy the leftmost two positions. The receive optical lanes occupy the rightmost two positions.

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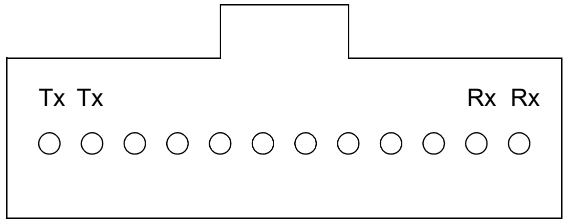

Figure 167–7—Optical lane assignments for 200GBASE-VR2 or 200GBASE-SR2

The four transmit and four receive optical lanes of 400GBASE-VR4 or 400GBASE-SR4 shall occupy the positions depicted in Figure 167–8 when looking into the MDI receptacle with the connector keyway feature on top. The interface contains eight active lanes within 12 total positions. The four center positions are unused. The transmit optical lanes occupy the leftmost four positions. The receive optical lanes occupy the rightmost four positions.

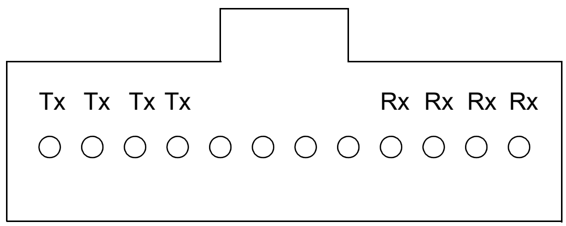

Figure 167–8—Optical lane assignments for 400GBASE-VR4 or 400GBASE-SR4

167.10.3.2 MDI requirements for 100GBASE-VR1 and 100GBASE-SR1

The MDI shall optically mate with the compatible plug on the optical fiber cabling. For 100GBASE-VR1 and 100GBASE-SR1, when the MDI is a duplex optical connector plug and receptacle connection, it shall meet the performance specifications of IEC 61753-1 and IEC 61753-022-2 (for category C environment) for performance grade Bm/2m. If the MDI uses a duplex optical connector it is not angled. If the MDI uses a multifiber connector it follows the requirements of 167.10.3.3.

167.10.3.3 MDI requirements for 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4

The MDI shall optically mate with the compatible plug on the optical fiber cabling.

For 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4 with a flat fiber interface the MDI adapter or receptacle shall meet the dimensional specifications for either interface 7-1-3: *MPO adapter interface—opposed keyway configuration* or interface 7-1-10: *MPO active device receptacle, flat interface*, as defined in IEC 61754-7-1. The plug terminating the optical fiber cabling shall meet the dimensional specifications of interface 7-1-4: *MPO female plug connector, flat interface for 2 to 12 fibre*s, as defined in IEC 61754-7-1. Figure 167–9 shows an MPO female plug connector with flat interface, and an MDI. The MDI connection shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-022-2 for performance grade Bm/2m.

NOTE—Flat fiber interfaces are the most commonly used for multifiber multimode systems.

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As an alternative, an optional angled fiber interface may be used for 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4. If the angled fiber interface is used, the MDI adapter or receptacle shall meet the dimensional specifications for either interface 7-1-3: *MPO adapter interface—opposed keyway configuration* or interface 7-1-9: *MPO active device receptacle, angled interface*, as defined in IEC 61754-7-1. The plug terminating the optical fiber cabling shall meet the dimensional specifications of interface 7-1-1: *MPO female plug connector, down-angled interface for 2 to 12 fibres*, as defined in IEC 61754-7-1. Figure 167–10 shows an MPO female plug connector with angled interface, and an MDI. The MDI connection shall meet the interface performance specifications of IEC 63267-1 for performance grade Bm/1m.[9]

A flat MDI adapter or receptacle is only compatible with a flat plug terminating the optical fiber cabling, and an angled MDI adapter or receptacle is only compatible with an angled plug terminating the optical fiber cabling.

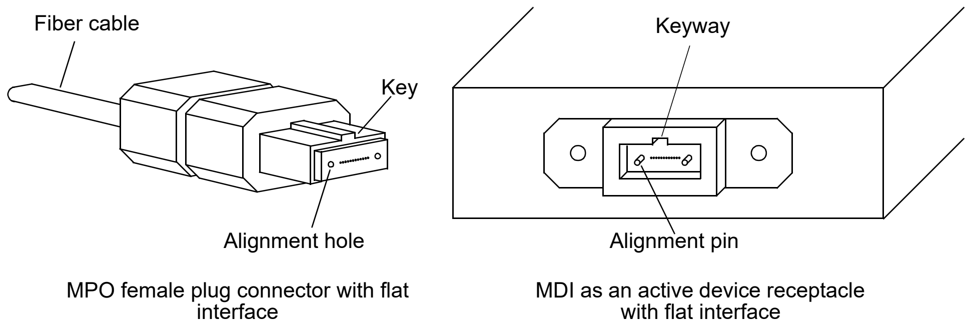

Figure 167–9—MPO female plug with flat interface and MDI active device receptacle with flat interface

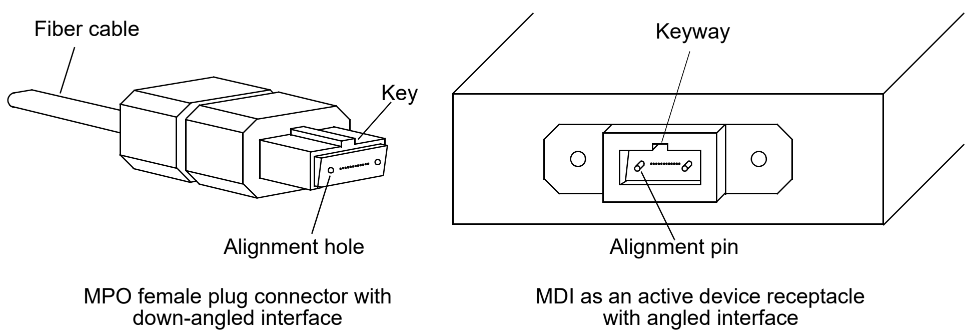

Figure 167–10—MPO female plug with down-angled interface and MDI active device receptacle with angled interface

[9]IEC 63267-1:2022 with performance grade 1m specification will be available in early 2023.

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167.11 Protocol implementation conformance statement (PICS) proforma for Clause 167, Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4[10]

167.11.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 167, Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

167.11.2 Identification

167.11.2.1 Implementation identification

Supplier[1]	
Contact point for inquiries about the PICS[1]	
Implementation Name(s) and Version(s)[1, 3]	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s)[2]	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

167.11.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3db-2022, Clause 167, Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [ ] Yes [ ] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3db-2022.)	

Date of Statement	

[10]*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

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167.11.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*VR1	100GBASE-VR1 PMD	167.7	Device supports requirements for 100GBASE-VR1 PHY	O	Yes [ ] No [ ]
*SR1	100GBASE-SR1 PMD	167.7	Device supports requirements for 100GBASE-SR1 PHY	O	Yes [ ] No [ ]
*VR2	200GBASE-VR2 PMD	167.7	Device supports requirements for 200GBASE-VR2 PHY	O	Yes [ ] No [ ]
*SR2	200GBASE-SR2 PMD	167.7	Device supports requirements for 200GBASE-SR2 PHY	O	Yes [ ] No [ ]
*VR4	400GBASE-VR4 PMD	167.7	Device supports requirements for 400GBASE-VR4 PHY	O	Yes [ ] No [ ]
*SR4	400GBASE-SR4 PMD	167.7	Device supports requirements for 400GBASE-SR4 PHY	O	Yes [ ] No [ ]
*INS	Installation / cable	167.10.1	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [ ] No [ ]
TP1	Reference point TP1 exposed and available for testing	167.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [ ] No [ ]
TP4	Reference point TP4 exposed and available for testing	167.5.1	This point may be made available for use by implementers to certify component conformance	O	Yes [ ] No [ ]
DC	Delay constraints	167.3.1	Device conforms to delay constraints	M	Yes [ ]
SC	Skew constraints	167.3.2	Device conforms to Skew and Skew Variation constraints	M	Yes [ ]
*MD	MDIO capability	167.4	Registers and interface supported	O	Yes [ ] No [ ]
*AFI	Angled fiber interface (multifiber connector only)	167.10.3.3		O	Yes [ ] No [ ]

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167.11.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4

167.11.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 100GBASE-R, 200GBASE-R, or 400GBASE-R PCS and PMA	167.1		M	Yes [ ]
F2	Integration with management functions	167.1		O	Yes [ ] No [ ]
F3	Bit error ratio	167.1.1	Meets the BER specified in 167.1.1	M	Yes [ ]
F4	Transmit function	167.5.2	Conveys bits from PMD service interface to MDI	M	Yes [ ]
F5	Mapping between optical signal and logical signal for transmitter	167.5.2	Optical power levels from lowest to highest correspond to tx_symbols zero, one, two, and three, respectively	M	Yes [ ]
F6	Receive function	167.5.3	Conveys symbols from MDI to PMD service interface	M	Yes [ ]
F7	Conversion of n optical signals to n electrical signals	167.5.3	For delivery to the PMD service interface	M	Yes [ ]
F8	Mapping between optical signal and logical signal for receiver	167.5.3	Optical power levels from lowest to highest correspond to rx_symbols zero, one, two, and three, respectively	M	Yes [ ]
F9	Global Signal Detect function	167.5.4	Report to the PMD service interface the message PMD:IS_SIGNAL.indication(SIGNAL_DETECT)	M	Yes [ ]
F10	Global Signal Detect behavior	167.5.4	SIGNAL_DETECT is a global indicator of the presence of optical signals on all n lanes	M	Yes [ ]
F11	Lane-by-lane Signal Detect function	167.5.5	Sets PMD_signal_detect_*i* values on a lane-by-lane basis per requirements of Table 167–5	MD:O	Yes [ ] No [ ] N/A [ ]
F12	PMD reset function	167.5.6	Resets the PMD sublayer	MD:O	Yes [ ] No [ ] N/A [ ]

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167.11.4.2 Management functions

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Management register set	167.4		MD:M	Yes [ ] N/A [ ]
M2	Global transmit disable function	167.5.7	Disables all of the optical transmitters with the PMD_global_transmit_disable variable	MD:O	Yes [ ] No [ ] N/A [ ]
M3	PMD lane-by-lane transmit disable function	167.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_*i* variable	MD:O	Yes [ ] No [ ] N/A [ ]
M4	PMD lane-by-lane transmit disable	167.5.8	Disables each optical transmitter independently if M3 = No	!MD:O	Yes [ ] No [ ] N/A [ ]
M5	PMD fault function	167.5.9	Sets PMD_fault to one if any local fault is detected	MD:O	Yes [ ] No [ ] N/A [ ]
M6	PMD transmit fault function	167.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	MD:O	Yes [ ] No [ ] N/A [ ]
M7	PMD receive fault function	167.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	MD:O	Yes [ ] No [ ] N/A [ ]

167.11.4.3 PMD to MDI optical specifications

Item	Feature	Subclause	Value/Comment	Status	Support
S1	Transmitter meets specifications in Table 167–7	167.7.1	Per definitions in 167.8	M	Yes [ ]
S2	Receiver meets specifications in Table 167–8	167.7.2	Per definitions in 167.8	M	Yes [ ]

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167.11.4.4 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable		2 m to 5 m in length	M	Yes [ ]
OM2	Center wavelength and spectral width	167.8.3	Per IEC 61280-1-3 under modulated conditions	M	Yes [ ]
OM3	Average optical power	167.8.4	Per IEC 61280-1-1	M	Yes [ ]
OM4	$\mathrm{OMA}_{\mathrm{outer}}$ measurements	167.8.5	Each lane	M	Yes [ ]
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	167.8.6	Each lane	M	Yes [ ]
OM6	Transmitter eye closure for PAM4 (TECQ)	167.8.7	Each lane	M	Yes [ ]
OM7	Overshoot/undershoot	167.8.8	Each lane	M	Yes [ ]
OM8	Transmitter power excursion	167.8.9	Each lane	M	Yes [ ]
OM9	Extinction ratio	167.8.10	Per IEC 61280-2-2	M	Yes [ ]
OM10	Transmitter transition time	167.8.11	Each lane	M	Yes [ ]
OM11	$\mathrm{RIN}_{14}\mathrm{OMA}$ measurement procedure	167.8.12	Each lane	M	Yes [ ]
OM12	Stressed receiver sensitivity	167.8.14	See 167.8.14	M	Yes [ ]

167.11.4.5 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	167.9.1	Conforms to J.2	M	Yes [ ]
ES2	Laser safety—IEC Hazard Level 1M	167.9.2	Conforms to Hazard Level 1M laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes [ ]
ES3	Electromagnetic interference	167.9.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes [ ]

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167.11.4.6 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	167.10	Meets requirements specified in Table 167–14	INS:M	Yes [ ] N/A [ ]
OC2	Optical fiber characteristics	167.10.1	Per Table 167–15	INS:M	Yes [ ] N/A [ ]
OC3	Maximum discrete reflectance	167.10.2.2	Less than –20 dB	INS:M	Yes [ ] N/A [ ]
OC4	MDI layout for 200GBASE-VR2 and 200GBASE-SR2	167.10.3.1	Optical lane assignments per Figure 167–7	(VR2 or SR2):M	Yes [ ] N/A [ ]
OC5	MDI layout for 400GBASE-VR4 and 400GBASE-SR4	167.10.3.1	Optical lane assignments per Figure 167–8	(VR4 or SR4):M	Yes [ ] N/A [ ]
OC6	MDI mating, 100GBASE-VR1 and 100GBASE-SR1, with duplex optical fiber connector	167.10.3.2	MDI optically mates with plug on the cabling, performance grade Bm/2m	(VR1 or SR1):M	Yes [ ] N/A [ ]
OC7	MDI requirements for 100GBASE-VR1 and 100GBASE-SR1, with duplex optical fiber connector	167.10.3.2	Per IEC 61753-1 and IEC 61753-022-2	(VR1 or SR1) and INS:M	Yes [ ] N/A [ ]
OC8	MDI mating, with multifiber connector	167.10.3.3	MDI optically mates with plug on the cabling, performance grade Bm/2m	(VR1, SR1, VR2, SR2, VR4, or SR4):!AFI:M	Yes [ ] N/A [ ]
OC9	MDI mating, with multifiber connector	167.10.3.3	MDI optically mates with plug on the cabling, performance grade Bm/1m	(VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M	Yes [ ] N/A [ ]
OC10	MDI dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-3 or interface 7-1-10	(VR1, SR1, VR2, SR2, VR4, or SR4):!AFI:M	Yes [ ] N/A [ ]
OC11	MDI dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-3 or interface 7-1-9	(VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M	Yes [ ] N/A [ ]
OC12	Cabling connector dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-4	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):!AFI:M	Yes [ ] N/A [ ]

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Item	Feature	Subclause	Value/Comment	Status	Support
OC13	Cabling connector dimensions, with multifiber connector	167.10.3.3	Per IEC 61754-7-1 interface 7-1-1	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M	Yes [ ] N/A [ ]
OC14	MDI requirements, with multifiber connector	167.10.3.3	Per IEC 61753-1 and IEC 61753-022-2, performance grade Bm/2m	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):!AFI:M	Yes [ ] N/A [ ]
OC15	MDI requirements, with multifiber connector	167.10.3.3	Per IEC 63267-1, performance grade Bm/1m	INS and (VR1, SR1, VR2, SR2, VR4, or SR4):AFI:M	Yes [ ] N/A [ ]

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